

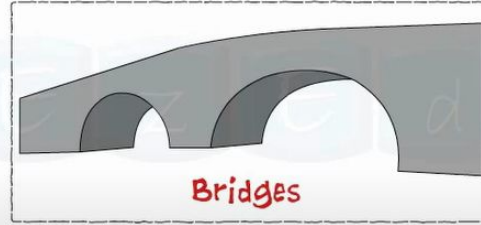
Engineering Curves

Engineering Curves:

- Ellipse
- Parabola
- Hyperbola
- Cycloid
- Epicycloid
- Hypocycloid
- Involute
- Spiral
- Helix etc

Engineering Curves:

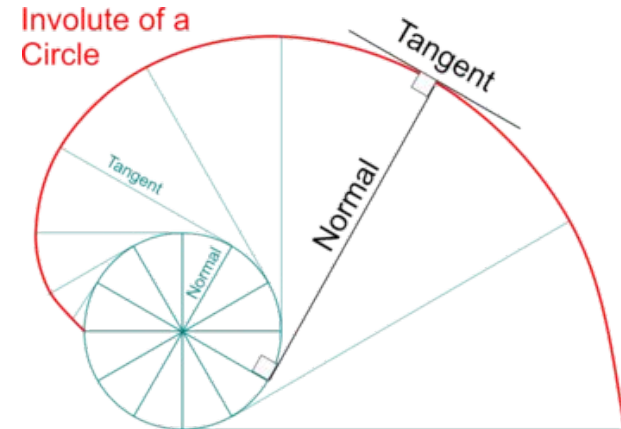
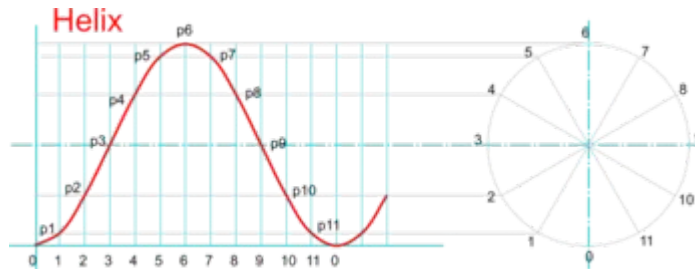
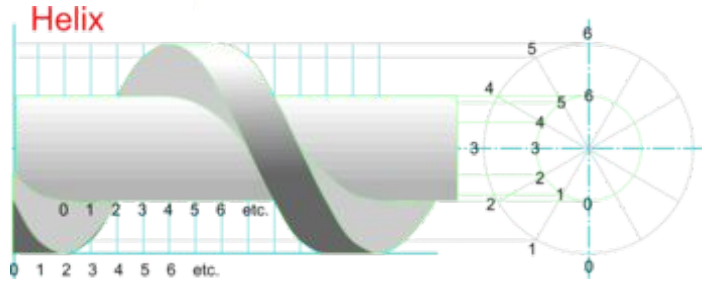
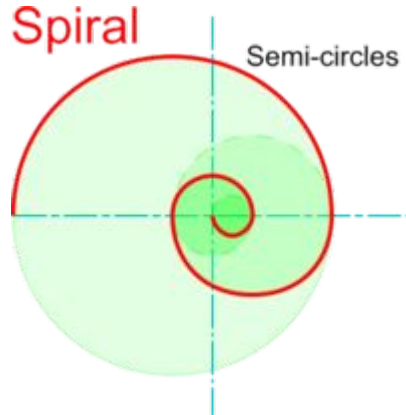
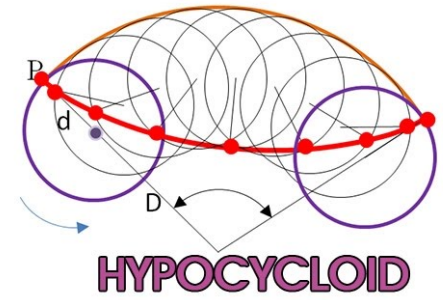
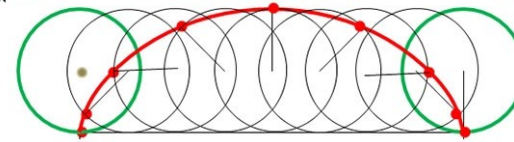
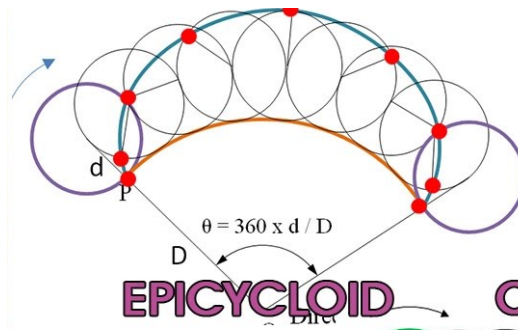
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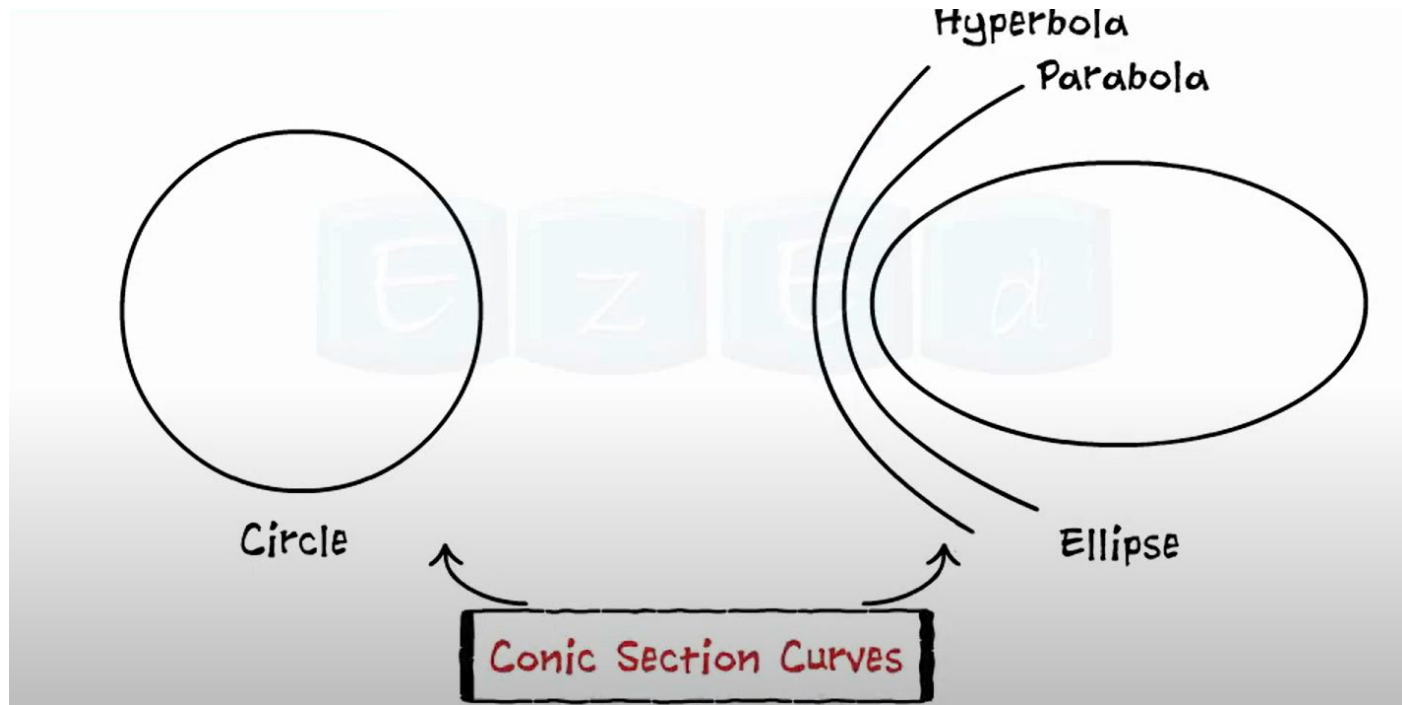
Use of curves

Engineering Curves:

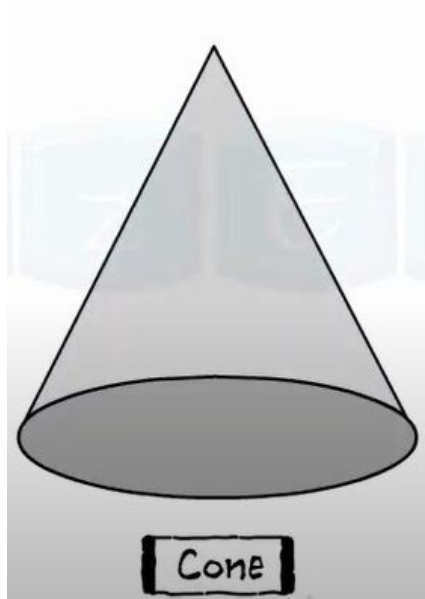
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Engineering Curves:

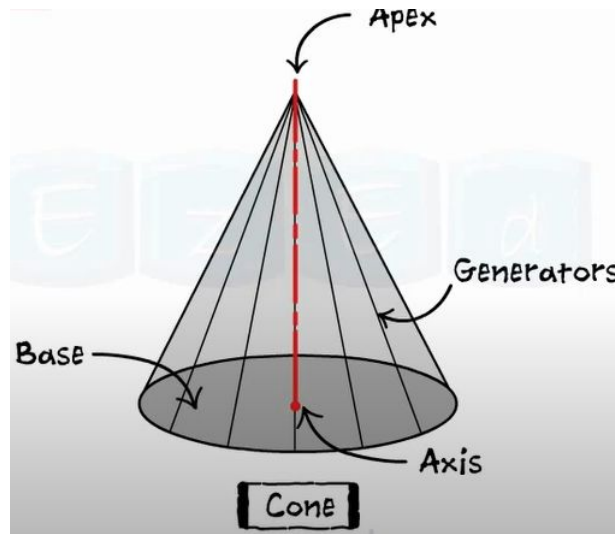


Engineering Curves:



Conic section curves are generated taking various sections of cone.

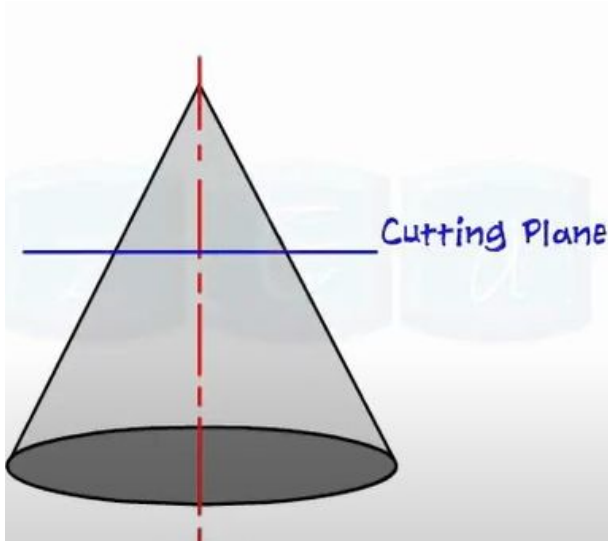
Engineering Curves:



Conic section curves are generated taking various sections of cone.

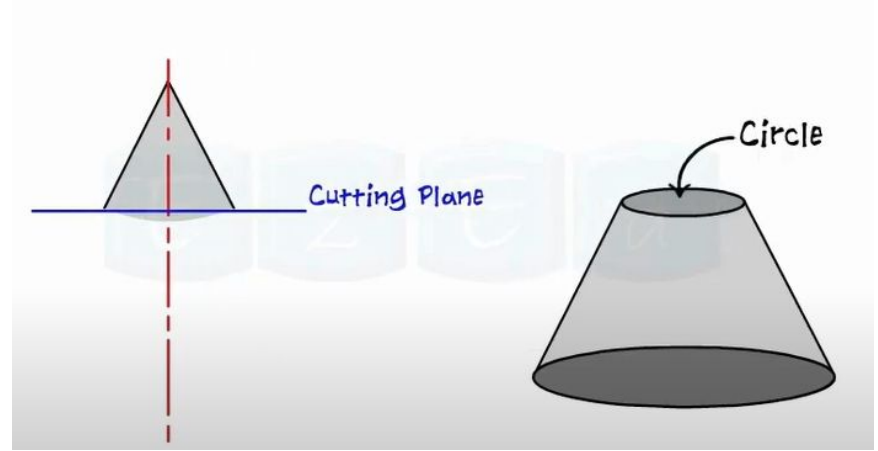
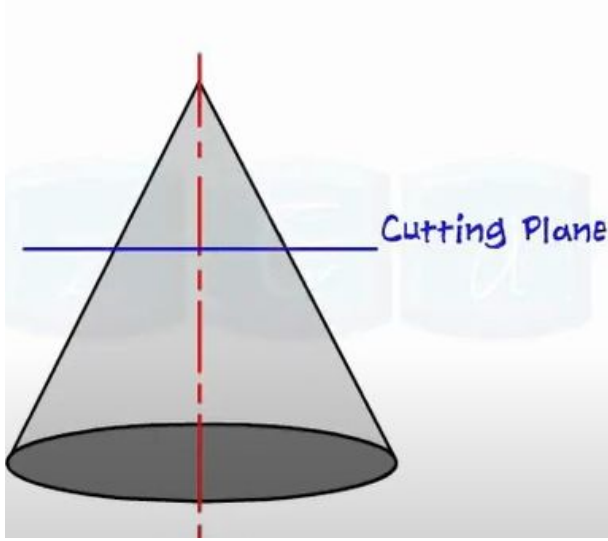
Engineering Curves (Circle):

Engineering Curves (Circle):



Cutting plane is perpendicular to the axis and cuts all the generators

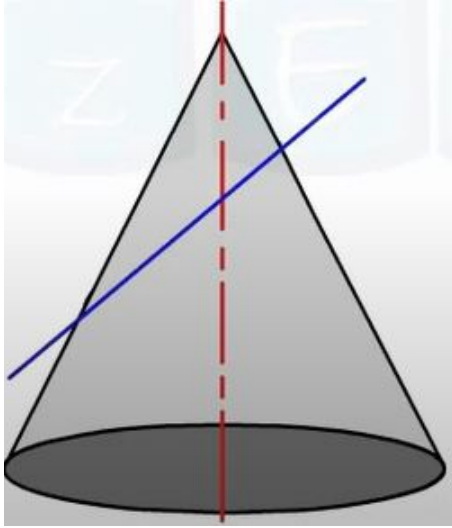
Engineering Curves (Circle):



Cutting plane is perpendicular to the axis and cuts all the generators

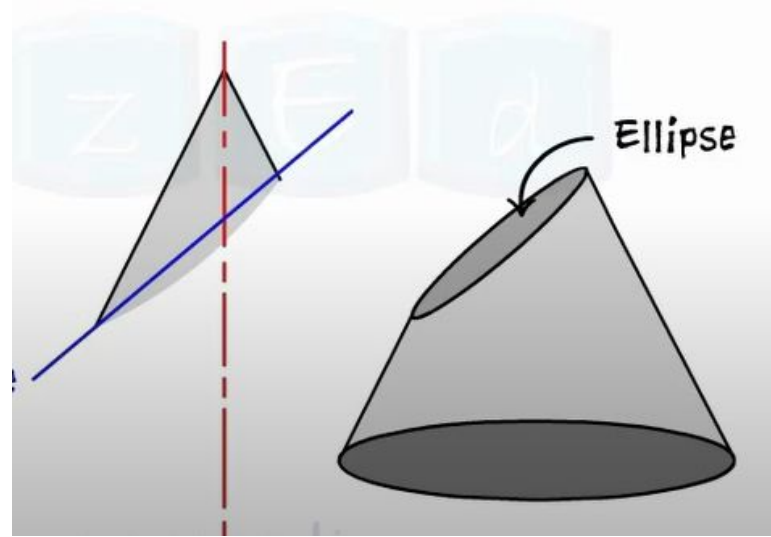
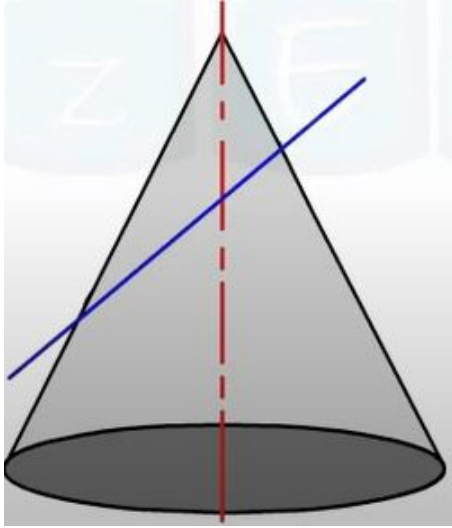
Engineering Curves (Ellipse):

Engineering Curves (Ellipse):



Cutting plane is inclined to the axis and cuts all the generators

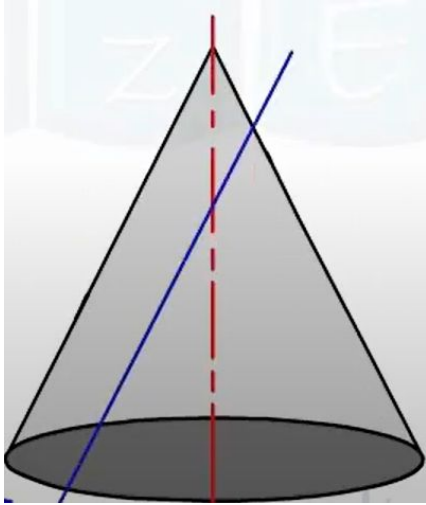
Engineering Curves (Ellipse):



Cutting plane is inclined to the axis and cuts all the generators

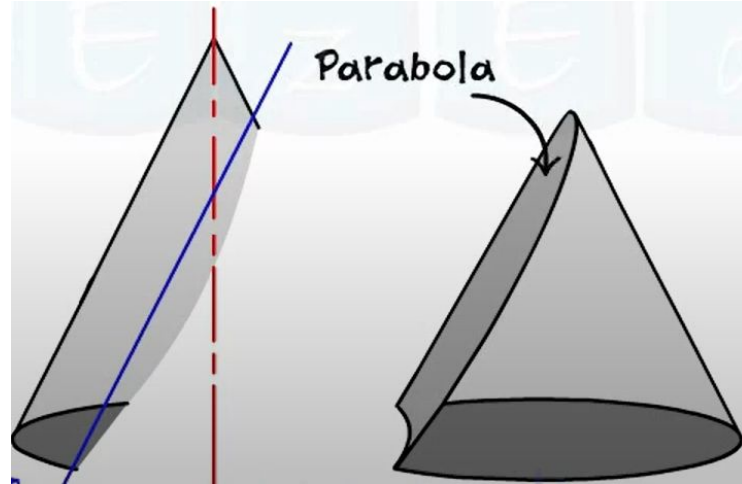
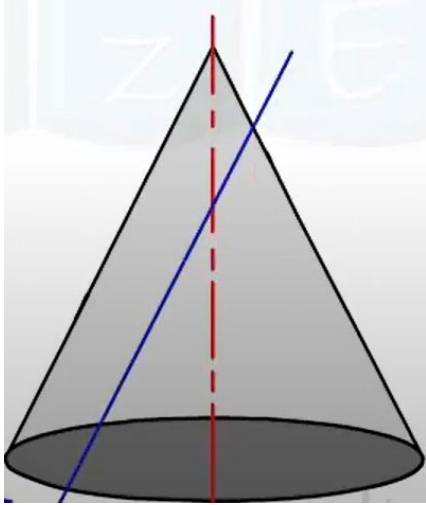
Engineering Curves (Parabola):

Engineering Curves (Parabola):



Cutting plane is inclined to the axis, and parallel to one of the generators

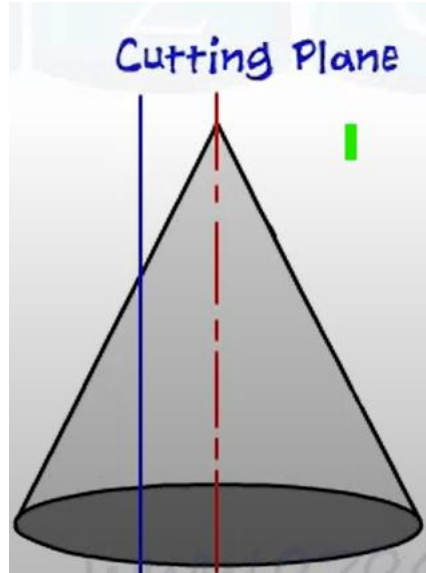
Engineering Curves (Parabola):



Cutting plane is inclined to the axis, and parallel to one of the generators

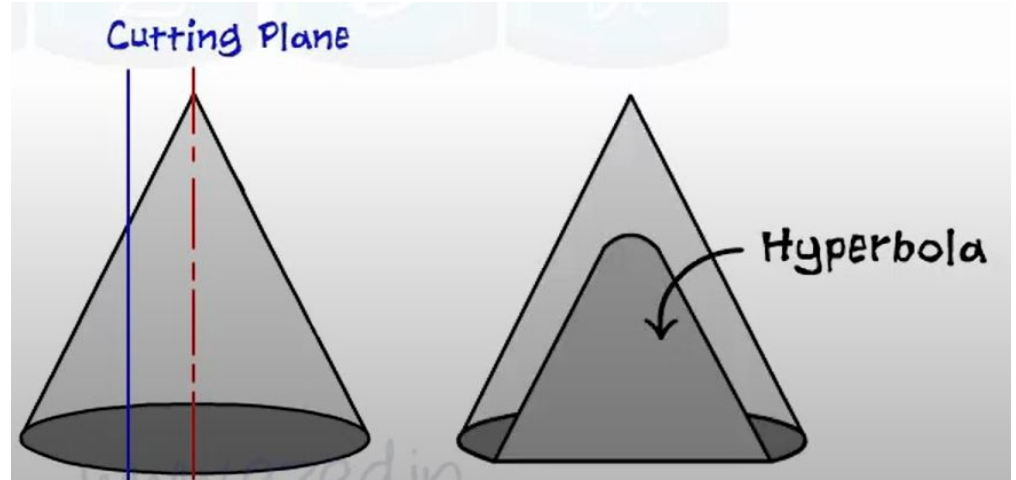
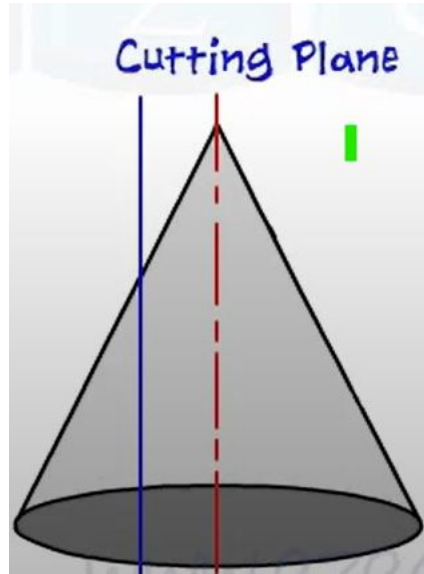
Engineering Curves (Hyperbola):

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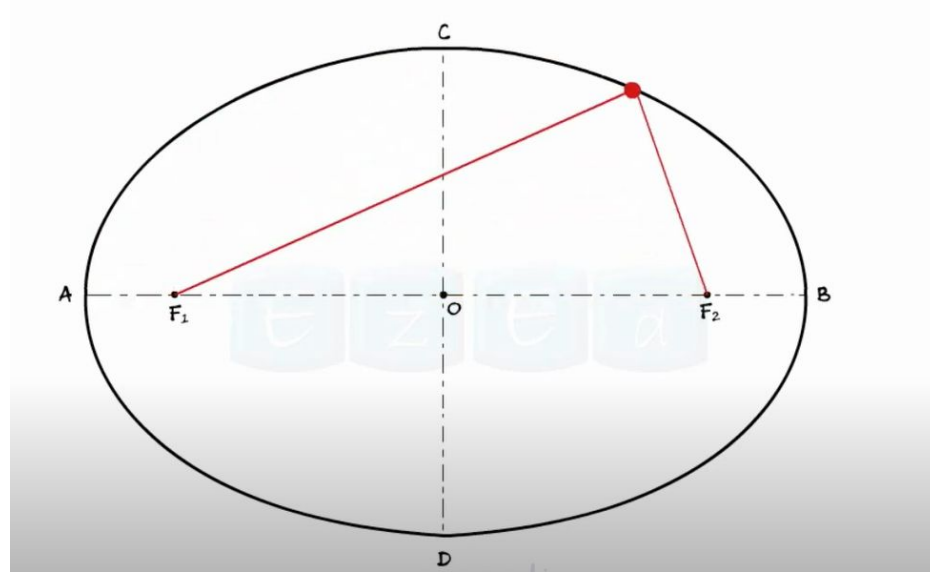
Cutting plane is parallel to the axis

Engineering Curves (Hyperbola):

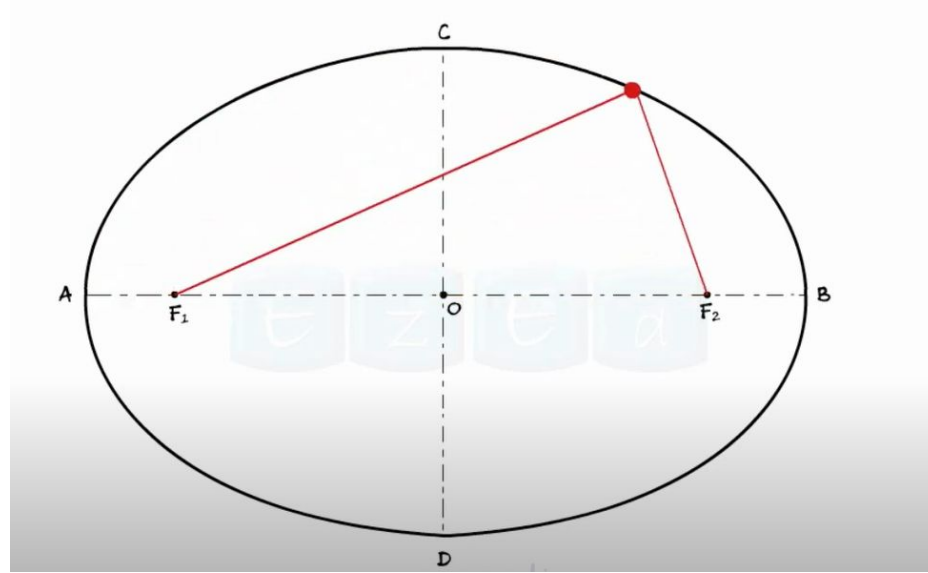


Cutting plane is parallel to the axis

Ellipse:

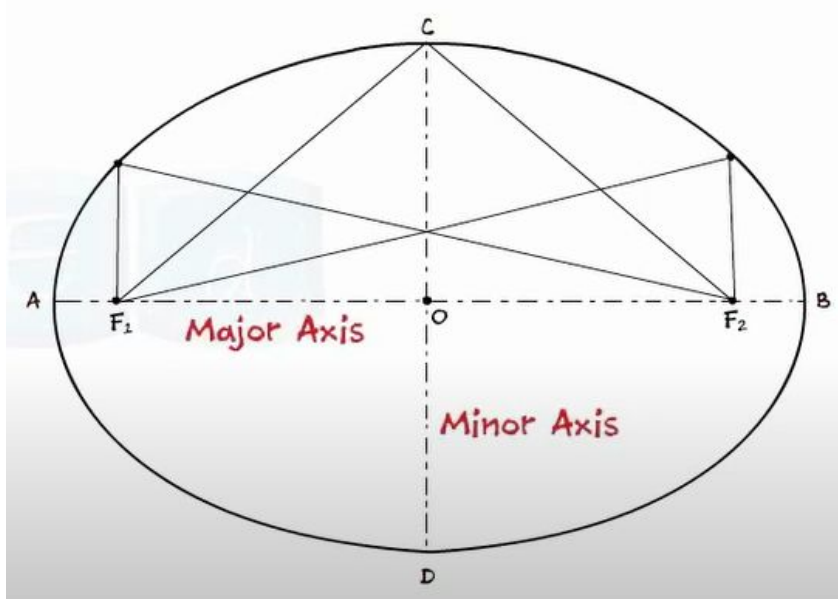


Ellipse:



An **ellipse** is defined as the set of all points in a plane such that **the sum of the distances from any point on the ellipse to two fixed points, called the foci (F_1 and F_2), is constant**

Ellipse:

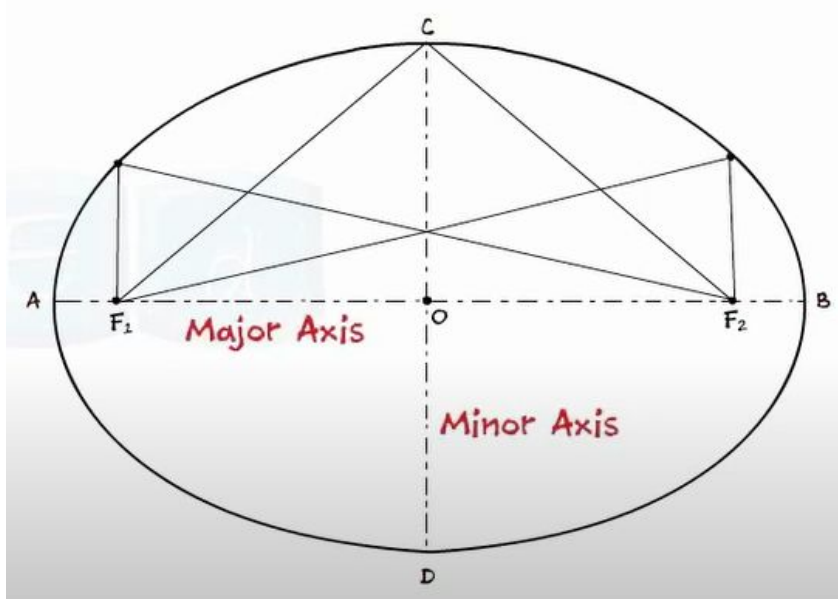


$$CF_1 + CF_2 = AB$$

$$CF_1 = \frac{1}{2} AB$$

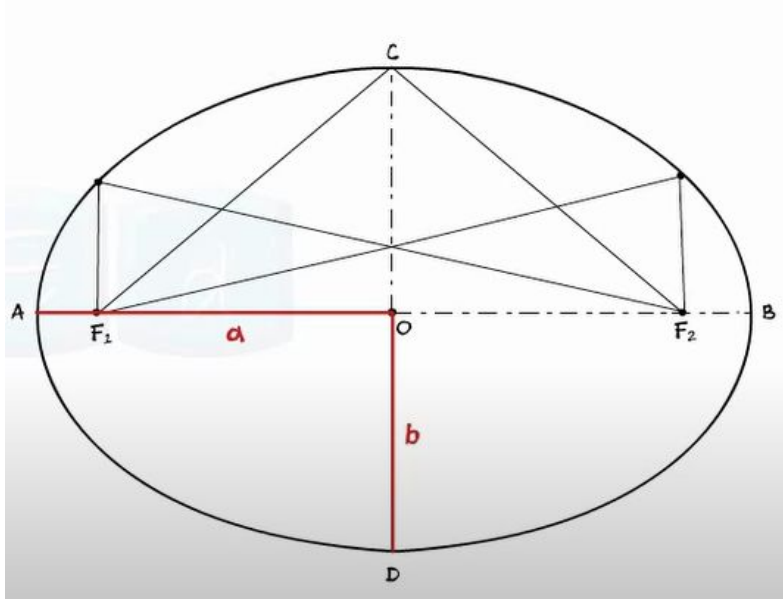
An **ellipse** is defined as the set of all points in a plane such that **the sum of the distances from any point on the ellipse to two fixed points, called the foci (F₁ and F₂), is constant**

Ellipse:



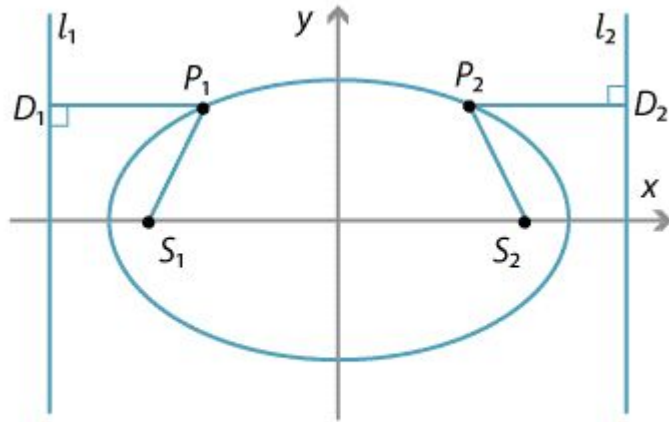
An **ellipse** is defined as the set of all points in a plane such that **the sum of the distances from any point on the ellipse to two fixed points, called the foci (F_1 and F_2), is constant**

Ellipse:



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Ellipse:



Focus-directrix definition of the ellipse.

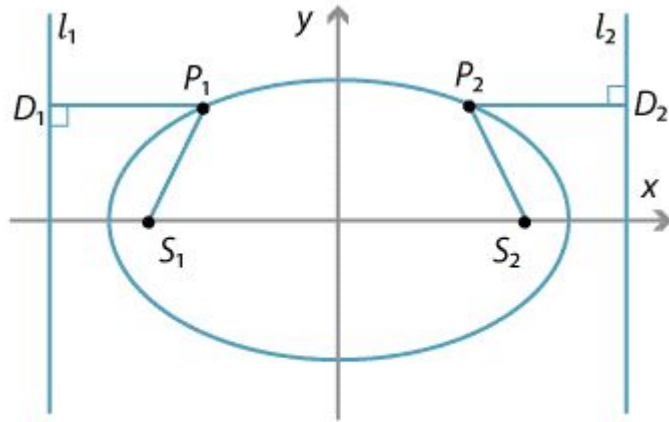
The **directrix** of an ellipse is a fixed line that is drawn outside the ellipse and is perpendicular to the major axis of the ellipse.

Eccentricity:

$$e = \frac{\text{distance of the point from the focus}}{\text{distance of the point from the directrix}}$$

Eccentricity is a measure of how much a curve, like a conic section, deviates from being a circle

Ellipse:



Focus-directrix definition of the ellipse.

$$e = \frac{\text{distance of the point from the focus}}{\text{distance of the point from the directrix}}$$

$$e < 1$$

Ways of Drawing an Ellipse

Ellipse

```
graph TD; A[Ellipse] --- B[Focus Directrix Method]; A --- C[Rectangle Method]; A --- D[Concentric Circle Method]; A --- E[Arcs of Circle Method];
```

Focus Directrix
Method

Rectangle
Method

Concentric Circle
Method

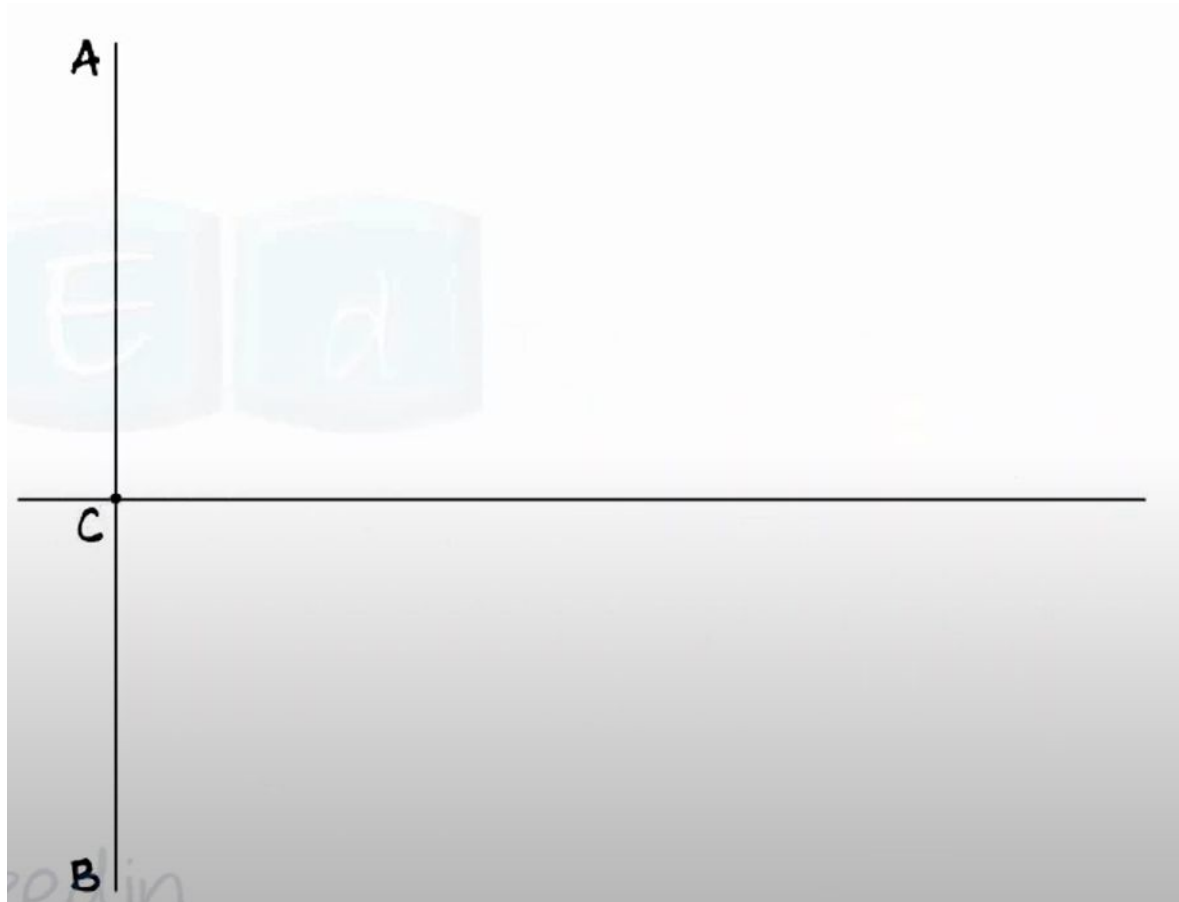
Arcs of Circle
Method

Problem:

Draw an **ellipse** when the distance of its focus from its directrix is 50mm and eccentricity is $2/3$. Also draw tangent and normal at any point Q on the curve using directrix.

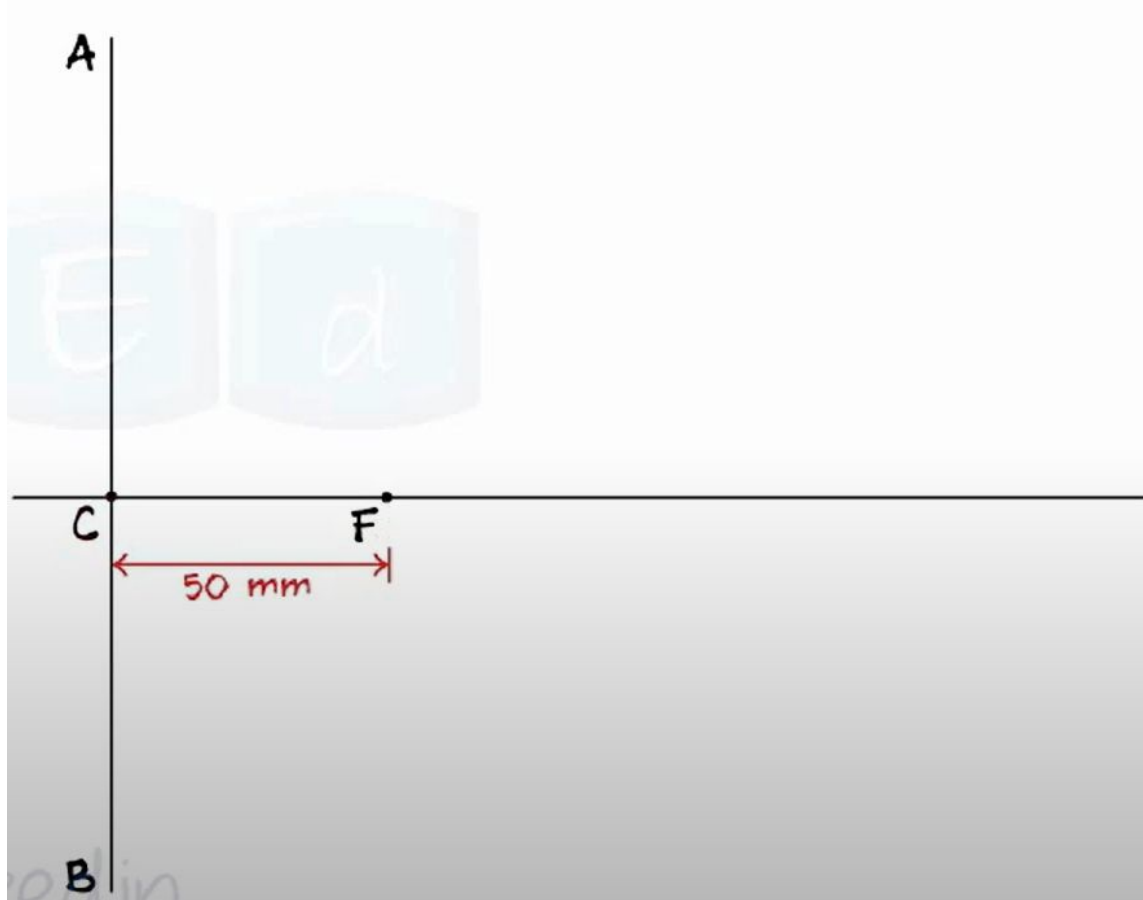
Ellipse using Focus Directrix Method:

Step 1: Draw a vertical line **AB** and a horizontal line from point **C**.



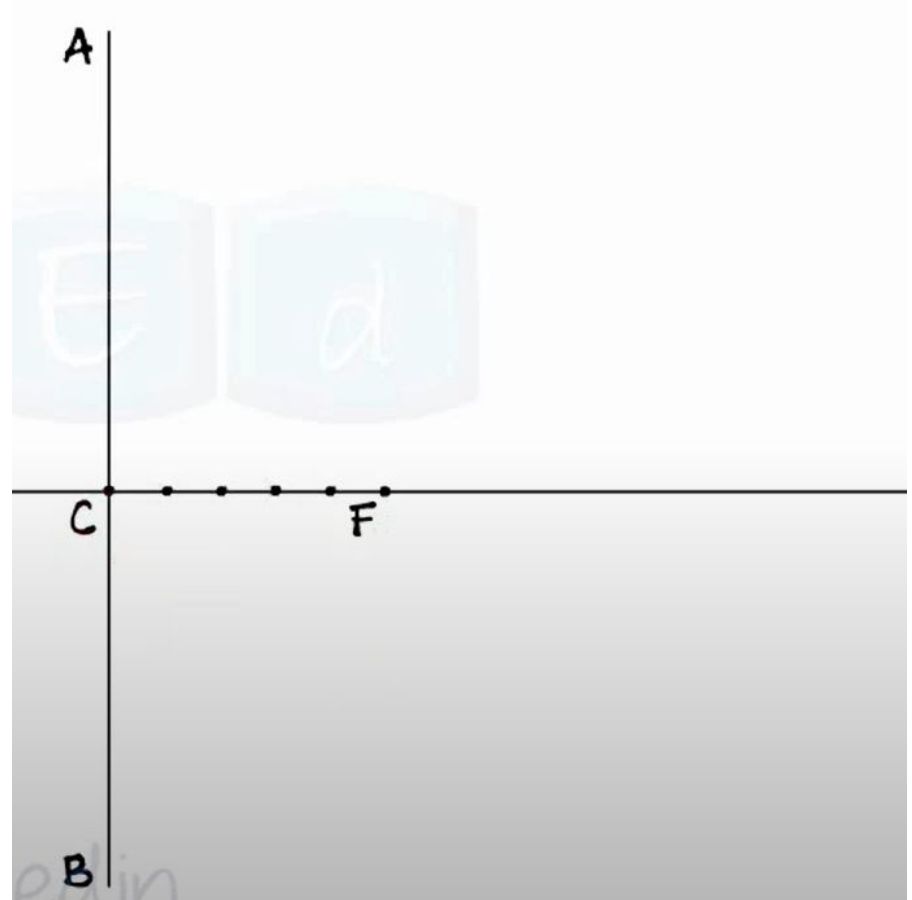
Ellipse using Focus Directrix Method:

Step 2: Mark the point **F** which is 50mm from point **C**.



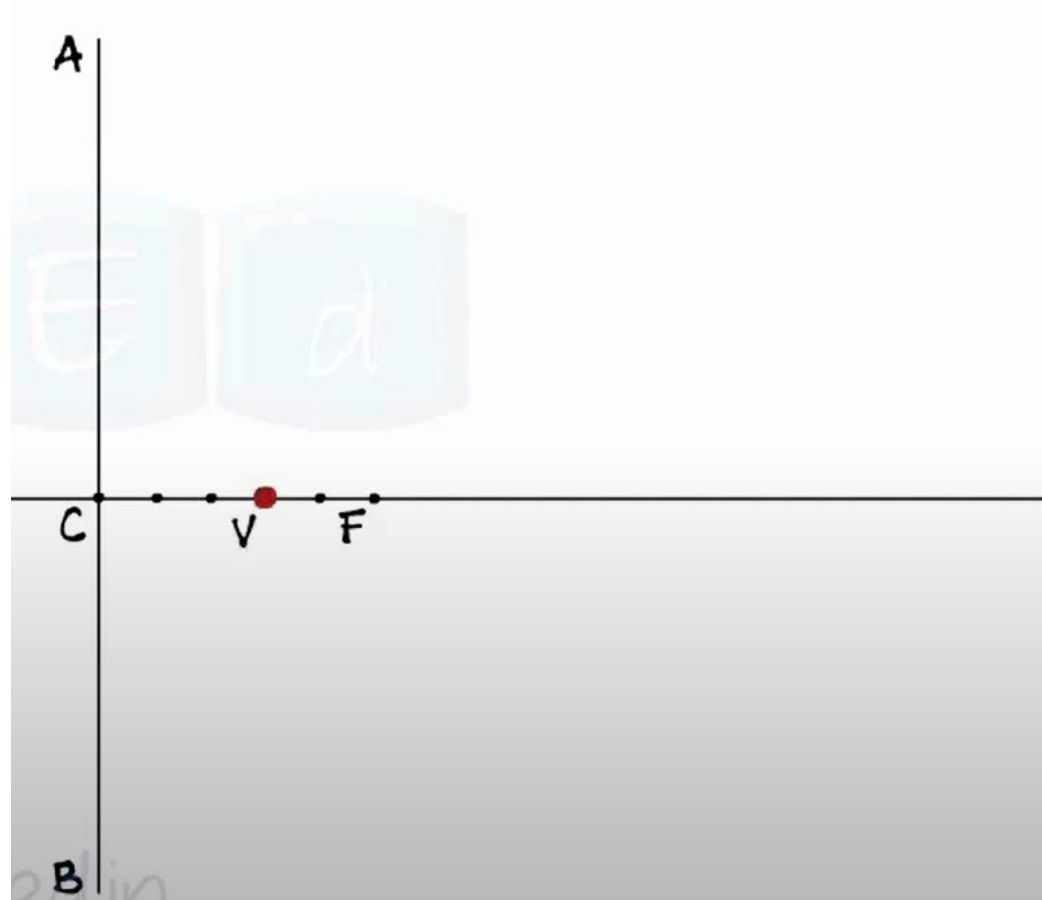
Ellipse using Focus Directrix Method:

Step 3: Divide the **CF** is 5 equal parts.



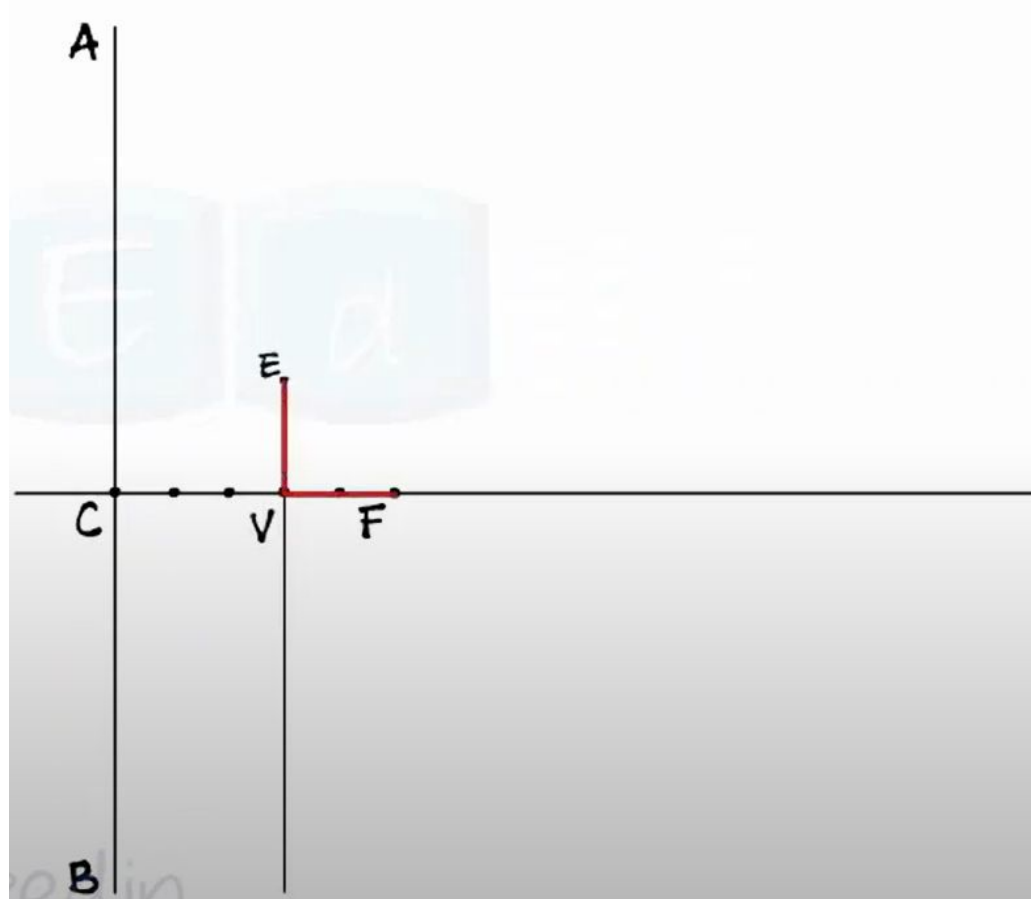
Ellipse using Focus Directrix Method:

Step 4: Mark the vertex V from the second division from F .



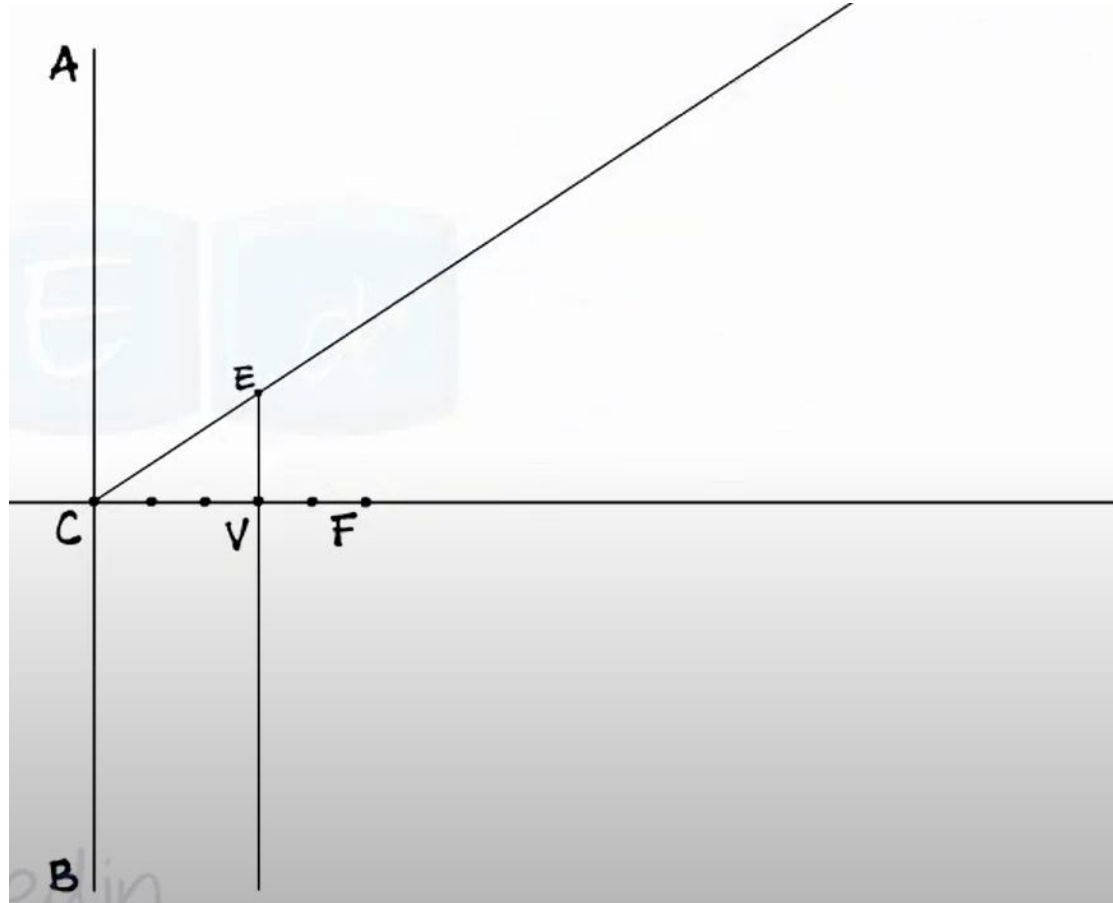
Ellipse using Focus Directrix Method:

Step 5: from V draw a perpendicular line **VE** such that $VE=VF$



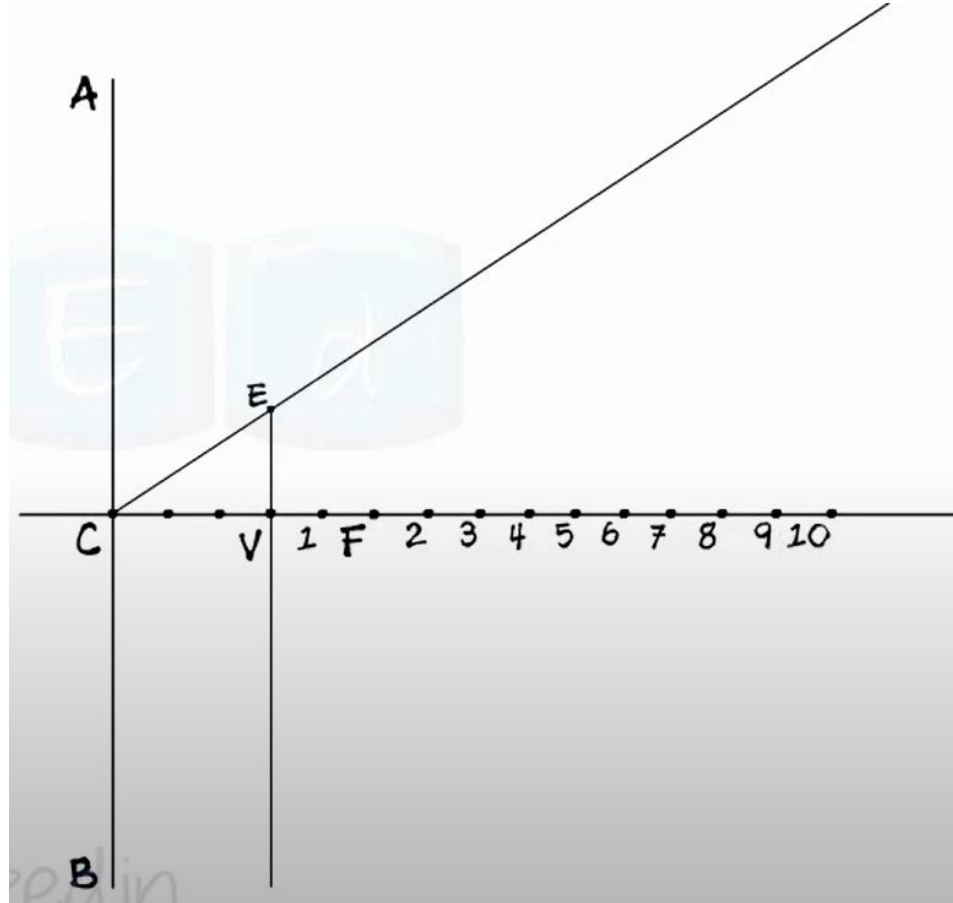
Ellipse using Focus Directrix Method:

Step 6: Join CE and extend it



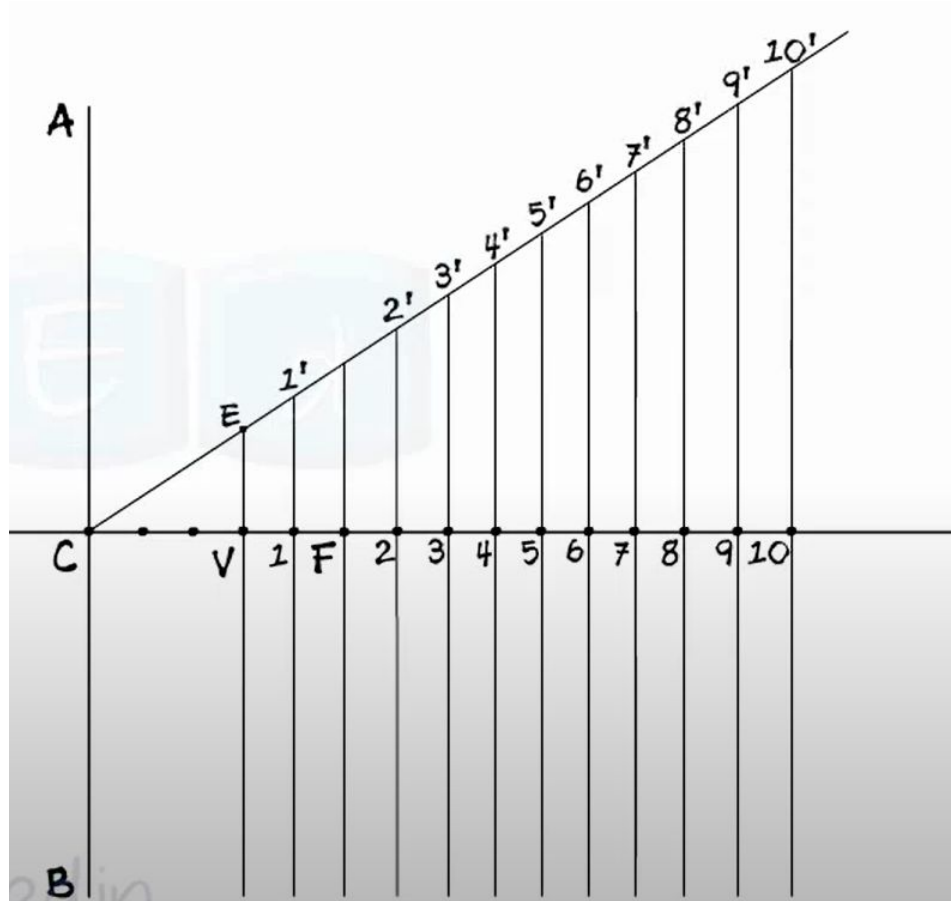
Ellipse using Focus Directrix Method:

Step 7: Divide the line into equal number of parts to the right of F.



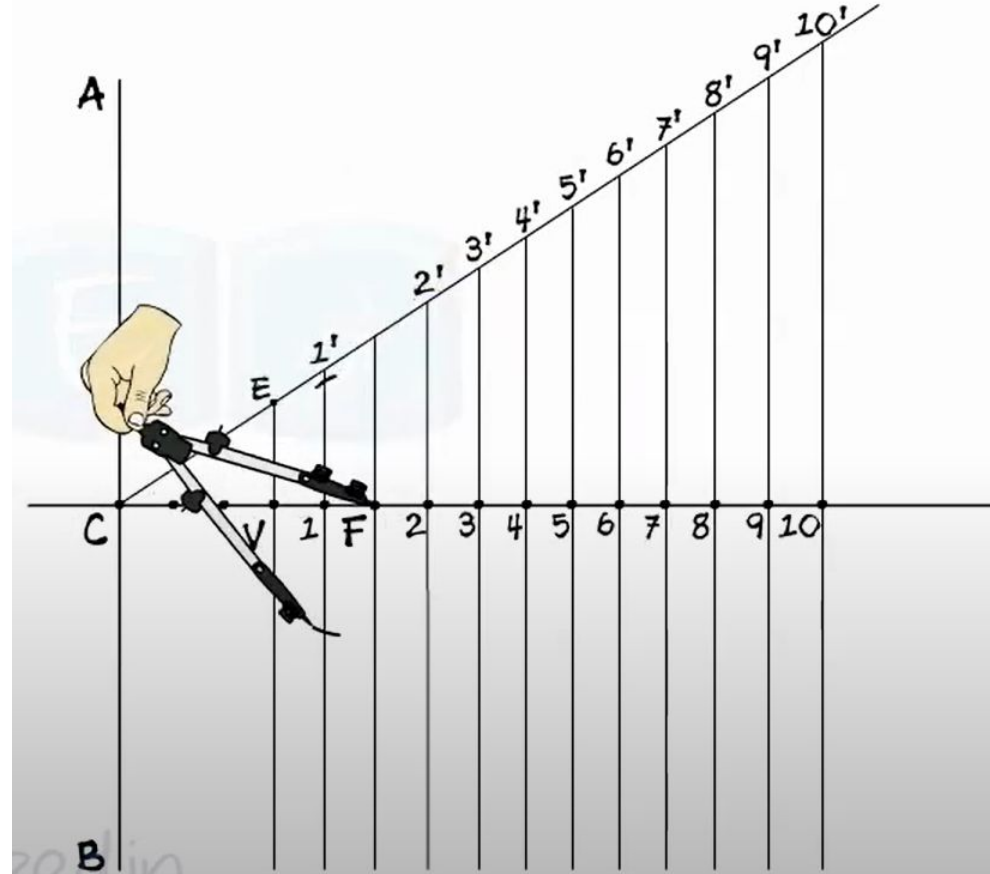
Ellipse using Focus Directrix Method:

Step 8: Draw perpendicular at these respective points.



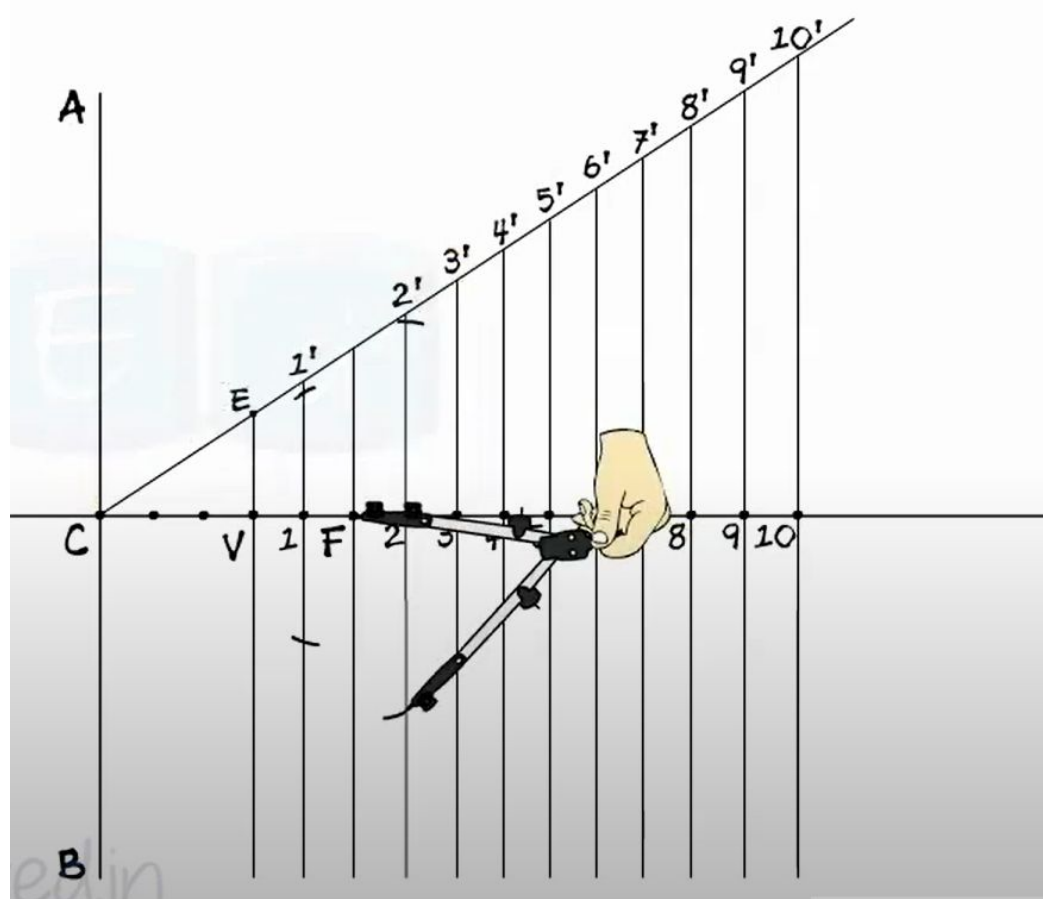
Ellipse using Focus Directrix Method:

Step 9: Measure the distance $11'$ and draw an arcs from point F to $11'$ on the either sides.



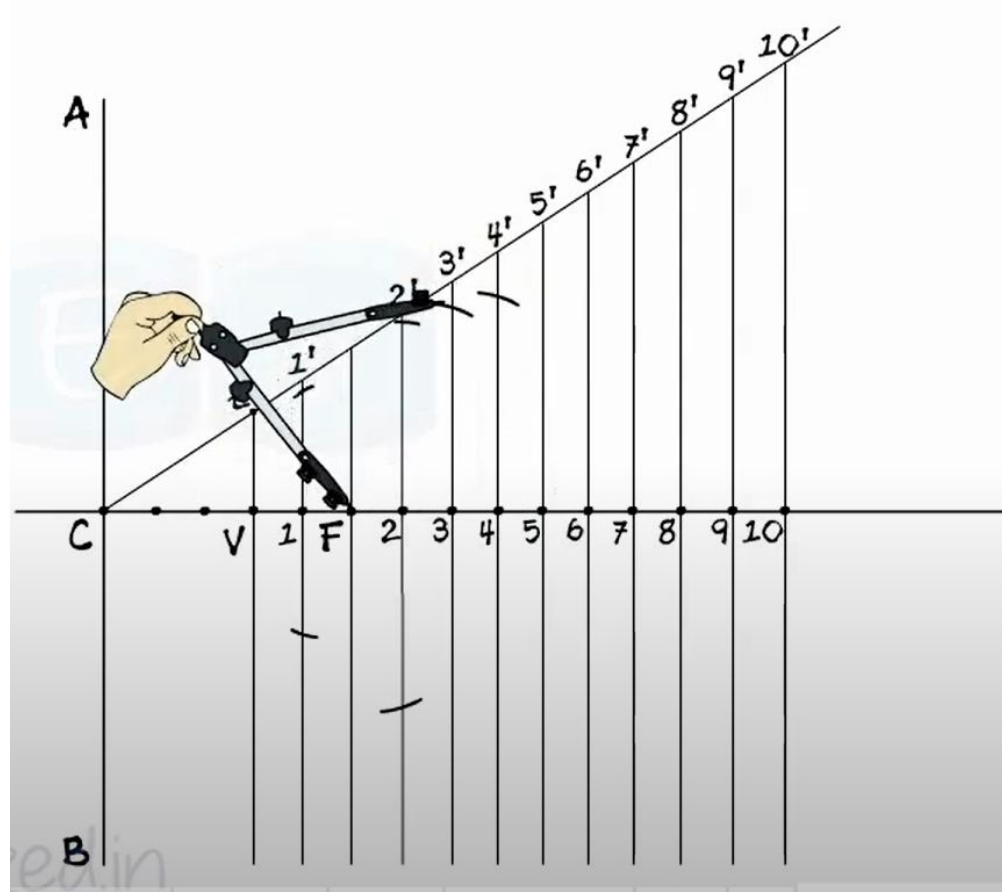
Ellipse using Focus Directrix Method:

Step 10: Similarly, measure the distance $22'$ and draw an arcs from point F to $22'$ on the either sides.



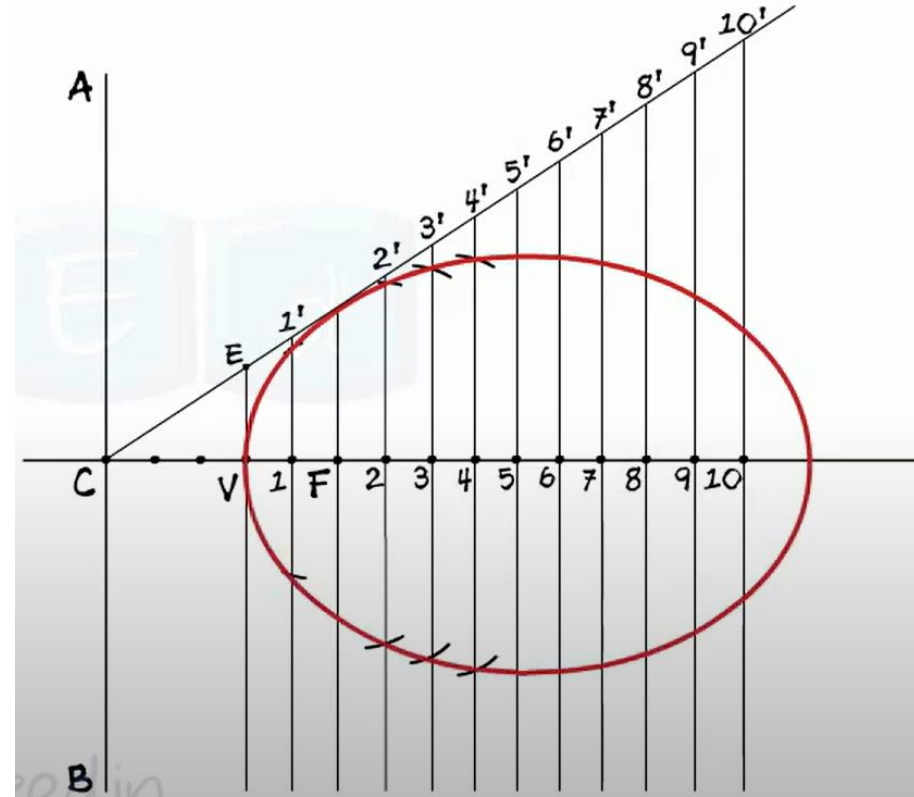
Ellipse using Focus Directrix Method:

Step 11: Continue this process for the remaining lines.



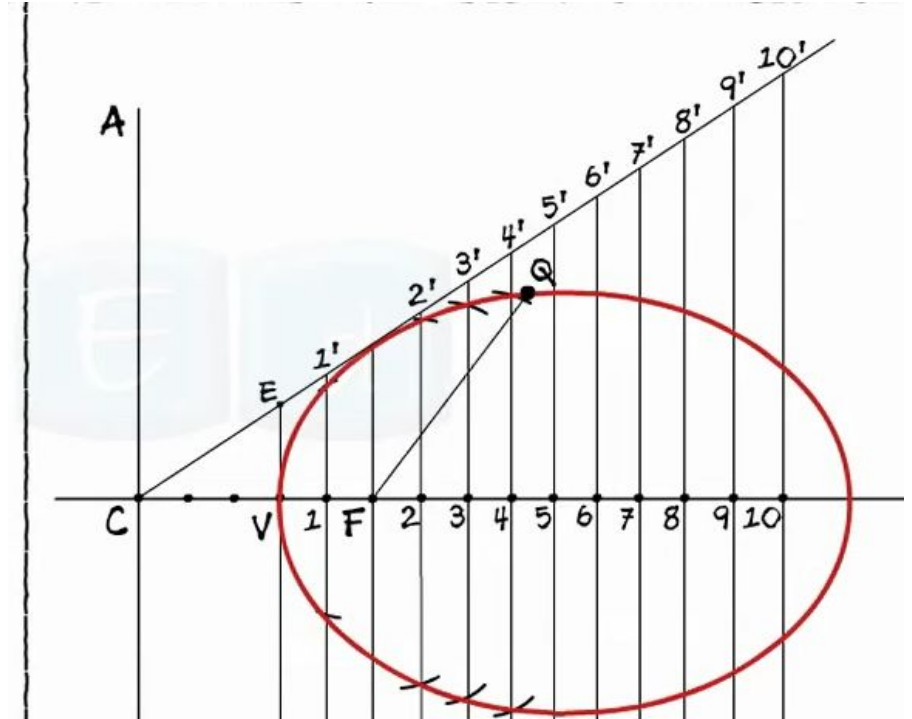
Ellipse using Focus Directrix Method:

Step 12: Join the points with a smooth curve. (Usually a french curve is used to join them)



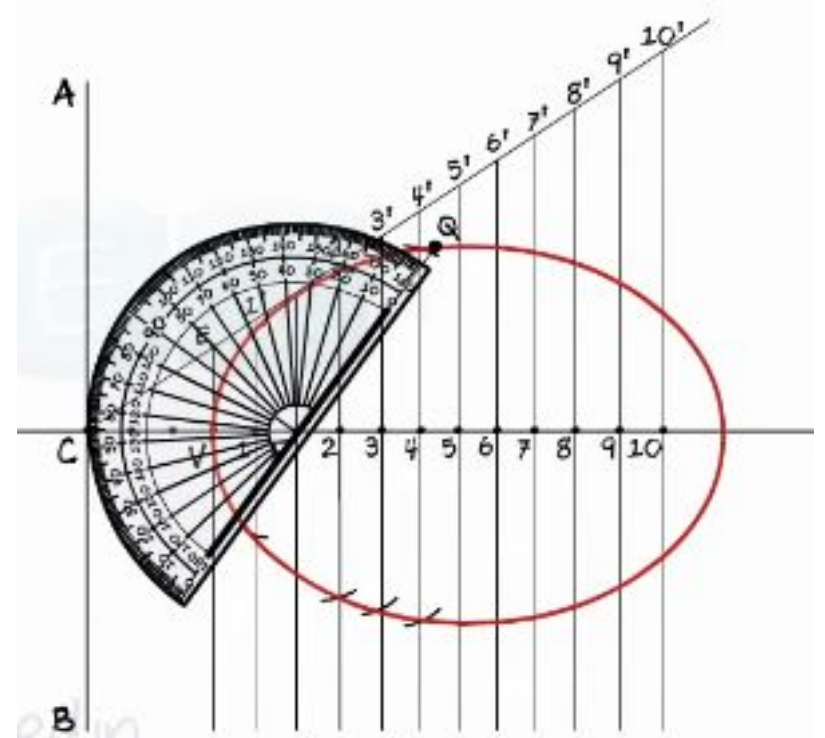
Ellipse using Focus Directrix Method:

Step 13: let us consider point Q anywhere in the ellipse join point Q to F



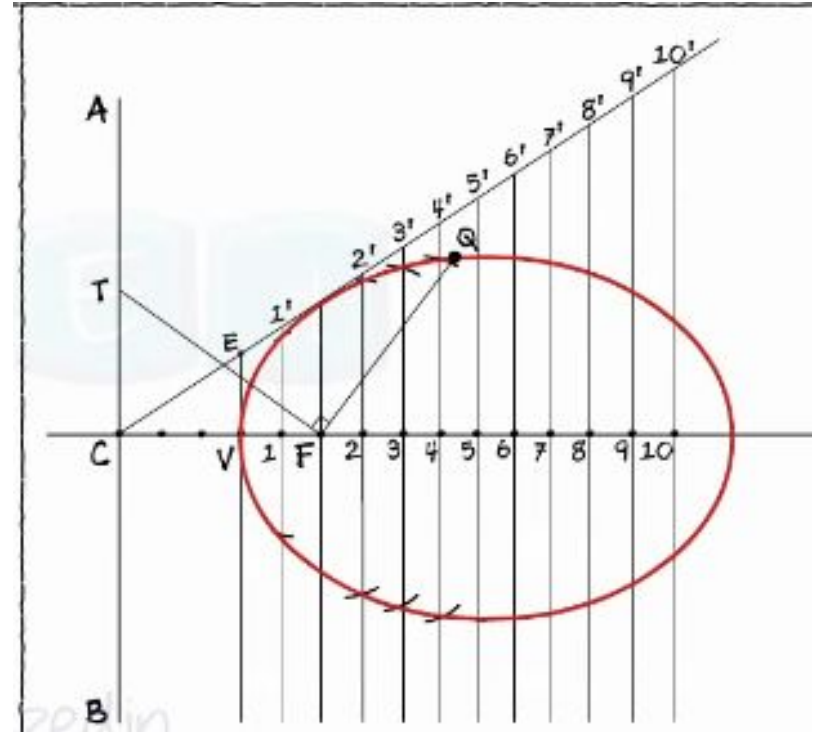
Ellipse using Focus Directrix Method:

Step 14: construct 90-degree angle with this line at point F



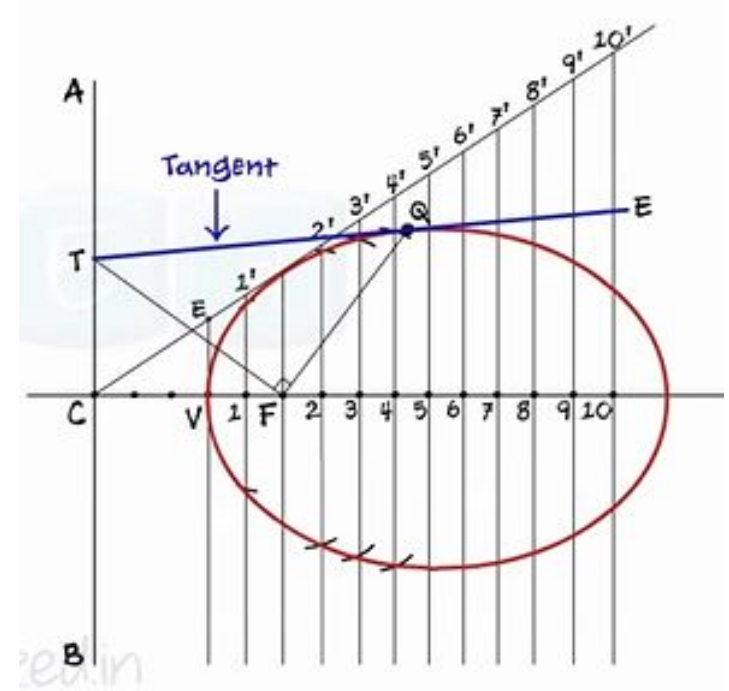
Ellipse using Focus Directrix Method:

Step 15: extend the line to meet directrix at T



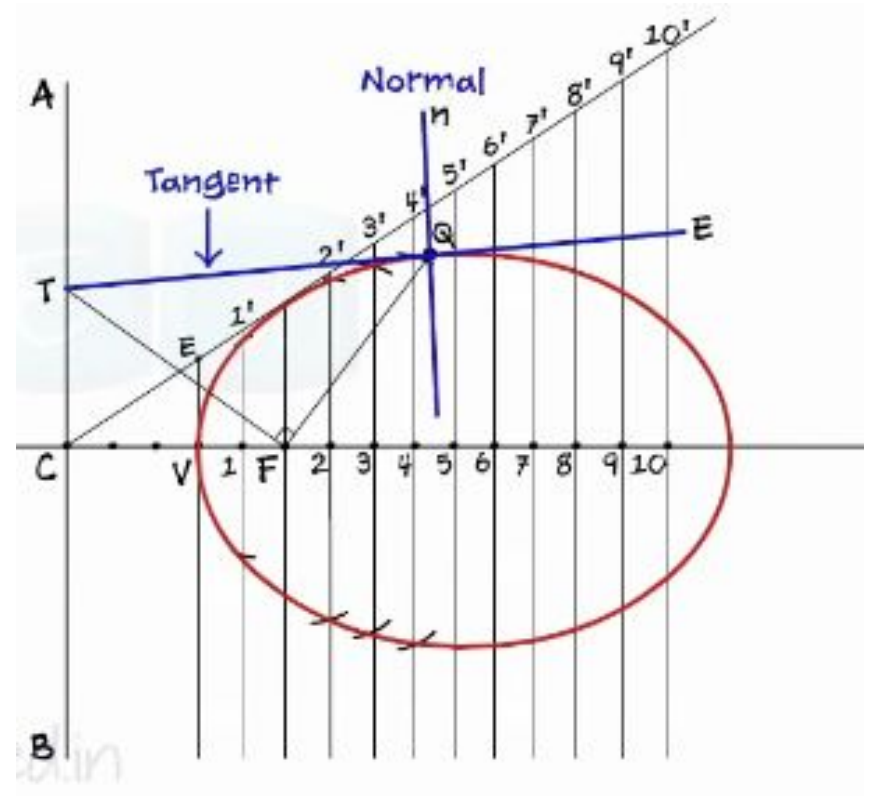
Ellipse using Focus Directrix Method:

Step 15: join this point to Q and extend it this line is tangent to the ellipse at Point Q



Ellipse using Focus Directrix Method:

Step 16: From point q draw a perpendicular line, this line is normal to the curve



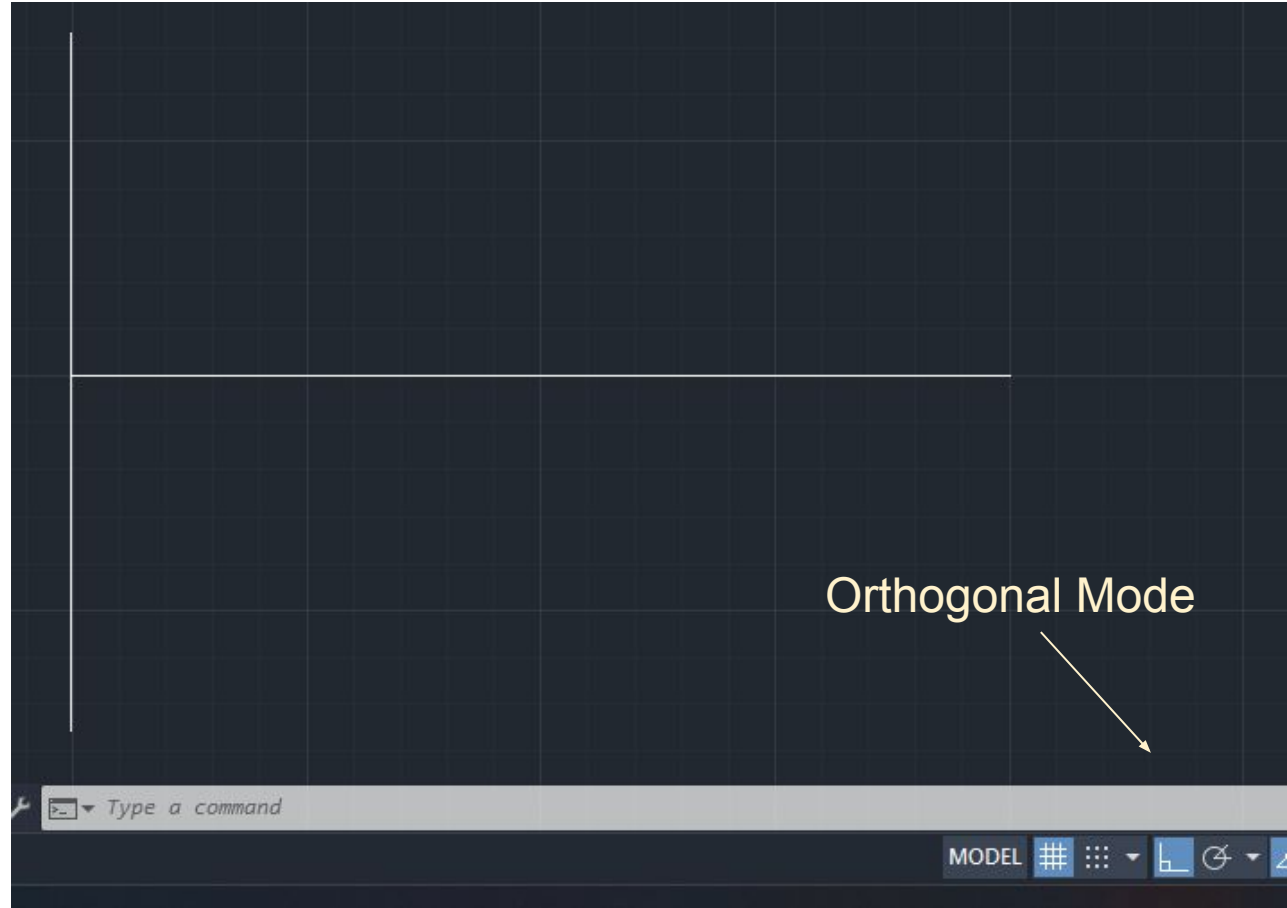


Drawing an Ellipse Using AutoCAD

Ellipse using Focus Directrix Method:

Step 1: We need to draw two lines (Directrix and Major Axis) perpendicular to each other.

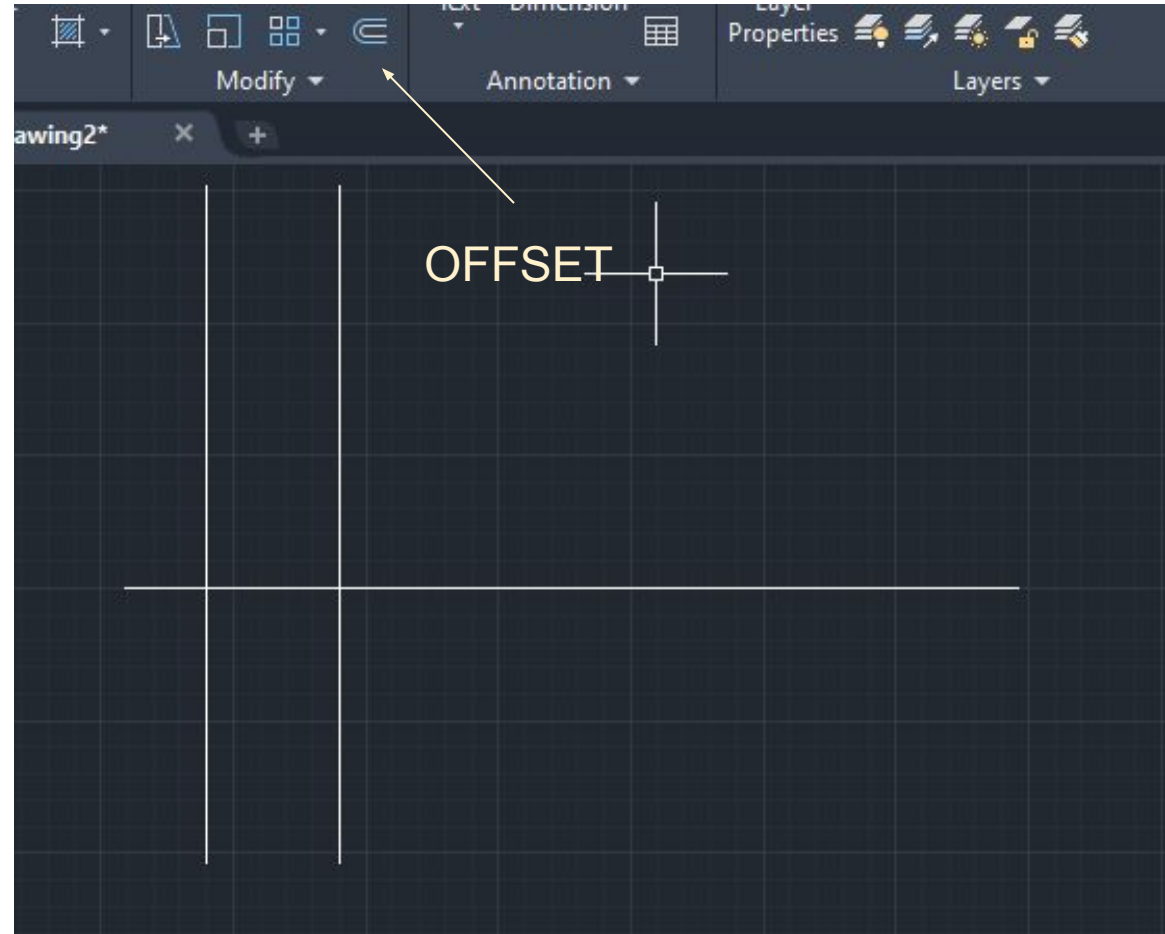
Note:
Turn on the orthogonal mode while drawing the perpendicular lines.



Ellipse using Focus Directrix Method:

Step 2: We need to find a point F which is 50mm from the directrix.

Note:
We will draw an *OFFSET* line at 50mm distance and also parallel to the directrix.



Ellipse using Focus Directrix Method:

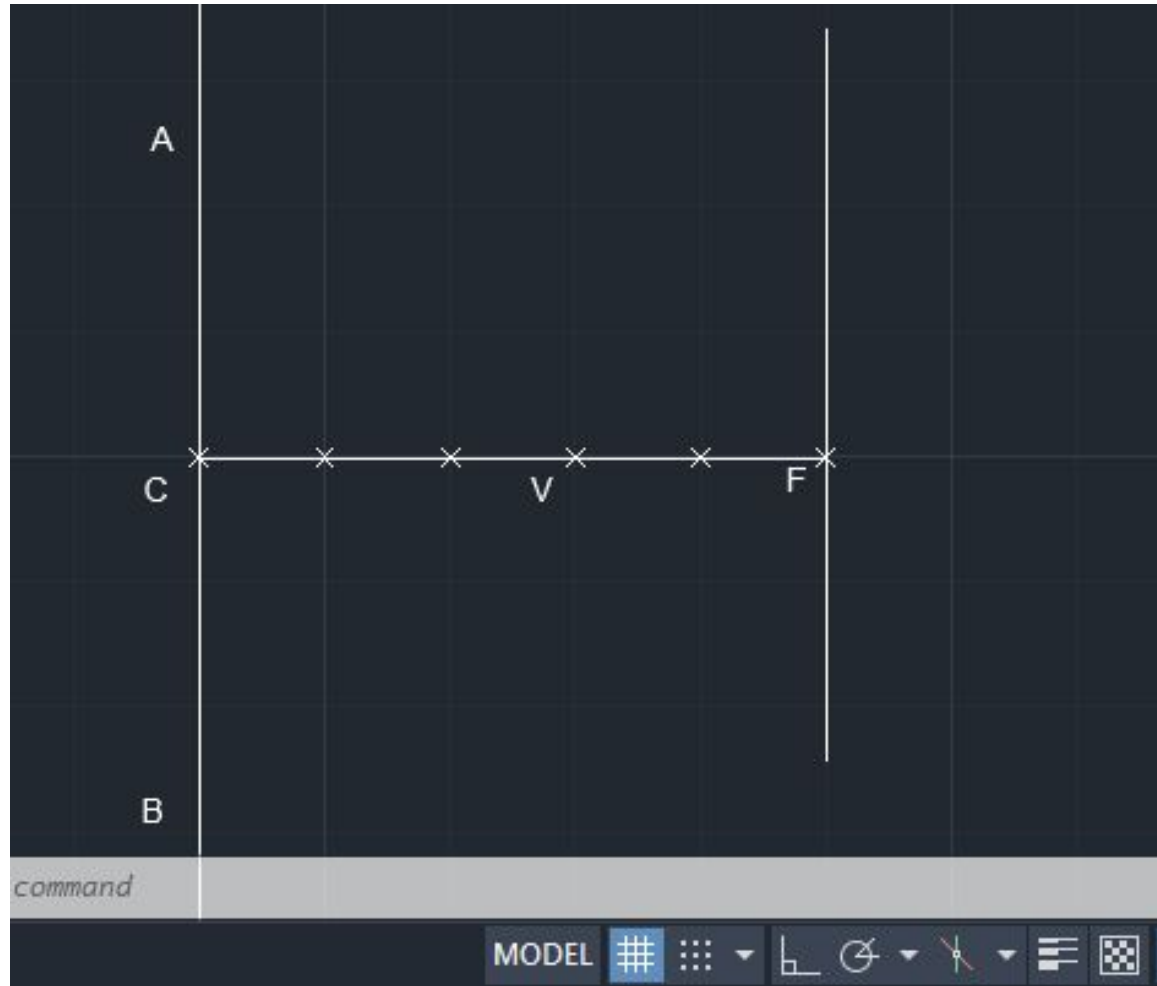
Step 3: TRIM extra lines.

DIVIDE CF into 5 equal parts and denote the second division point from F as V (vertex)

Note: (Node, Intersection from object snap)

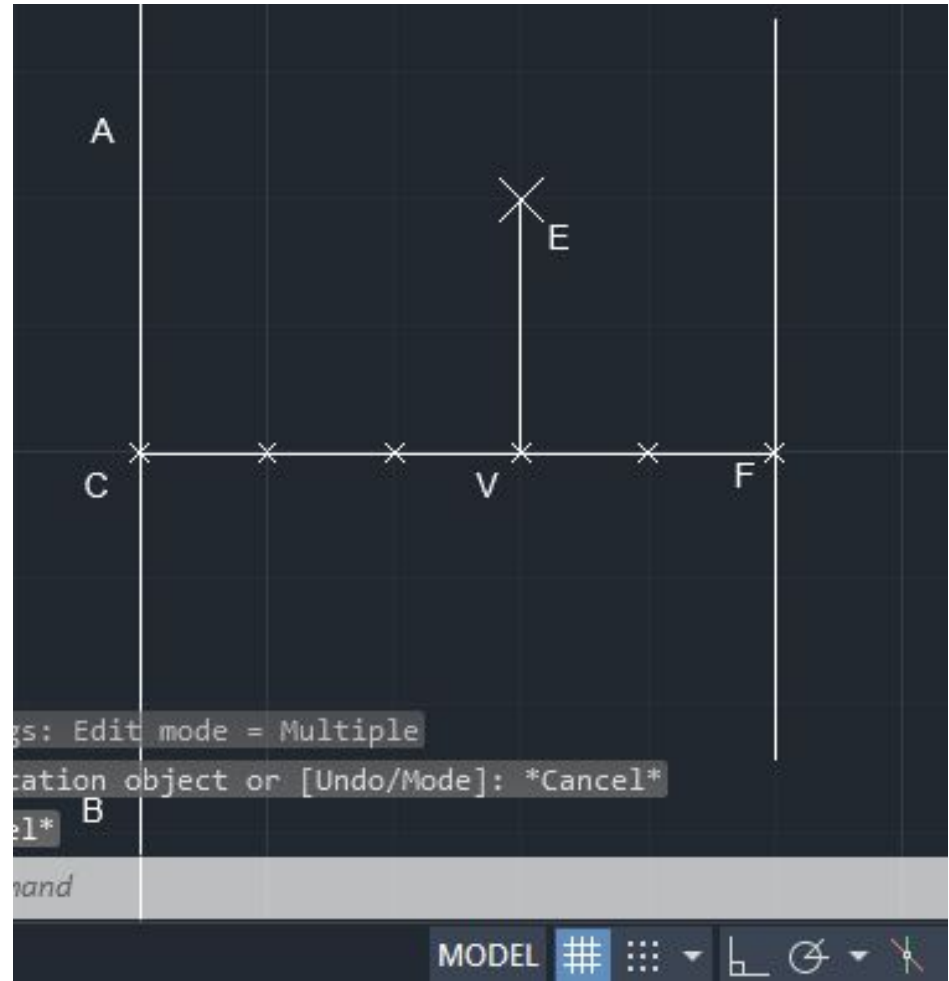
Commands you might need

- DDPTYPE (change the appearance and size of point objects in a drawing.)
- DIVIDE (a specified number of point objects or blocks at equal intervals along an existing object)
- POINT
- DT (TEXT)
- EX

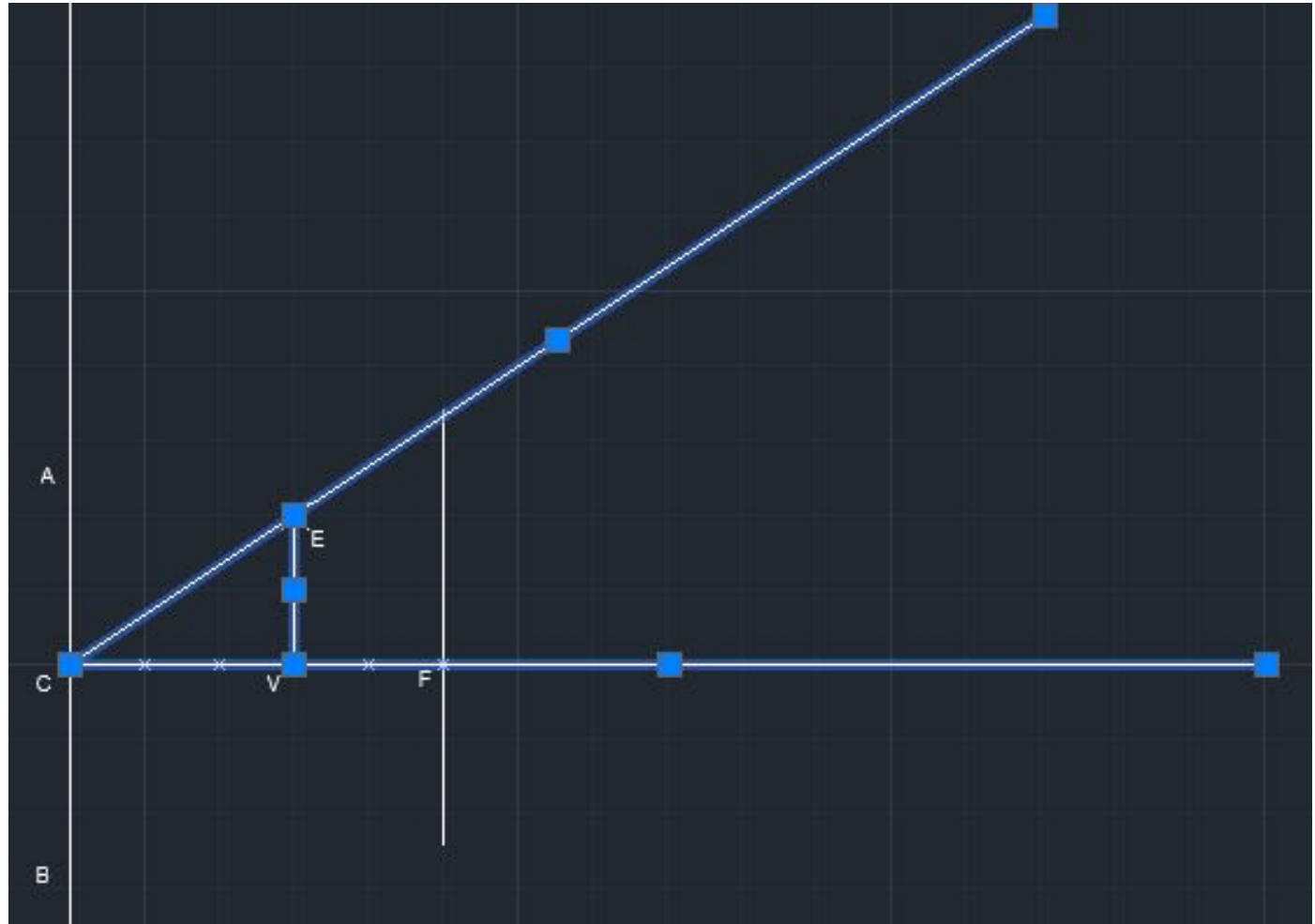


Ellipse using Focus Directrix Method:

Step 4: Then draw a perpendicular line VE at point V where $VE=VF$.



Step 5: Connect CE.
Extend CE and CF.

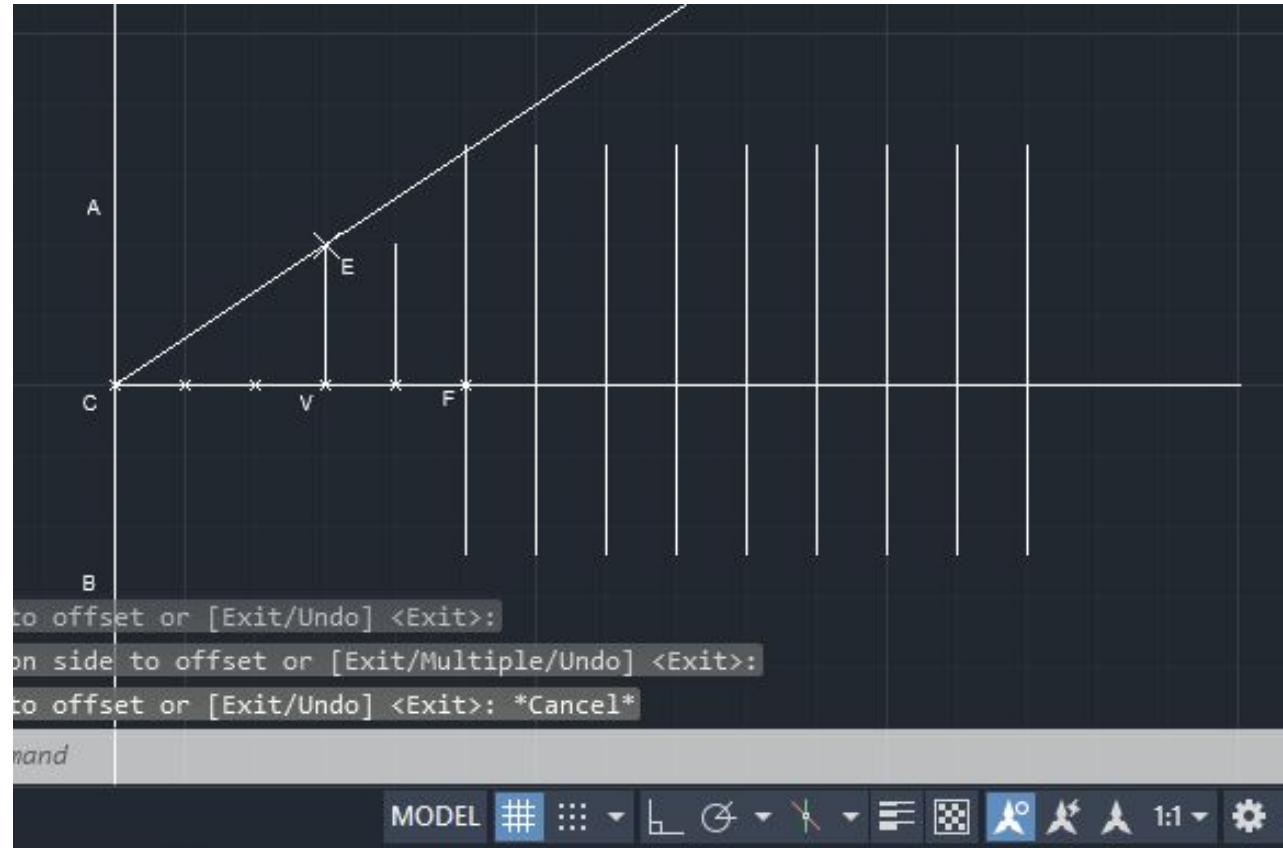


Ellipse using Focus Directrix Method:

Step 6: Draw few lines which are 10mm distant from each other.

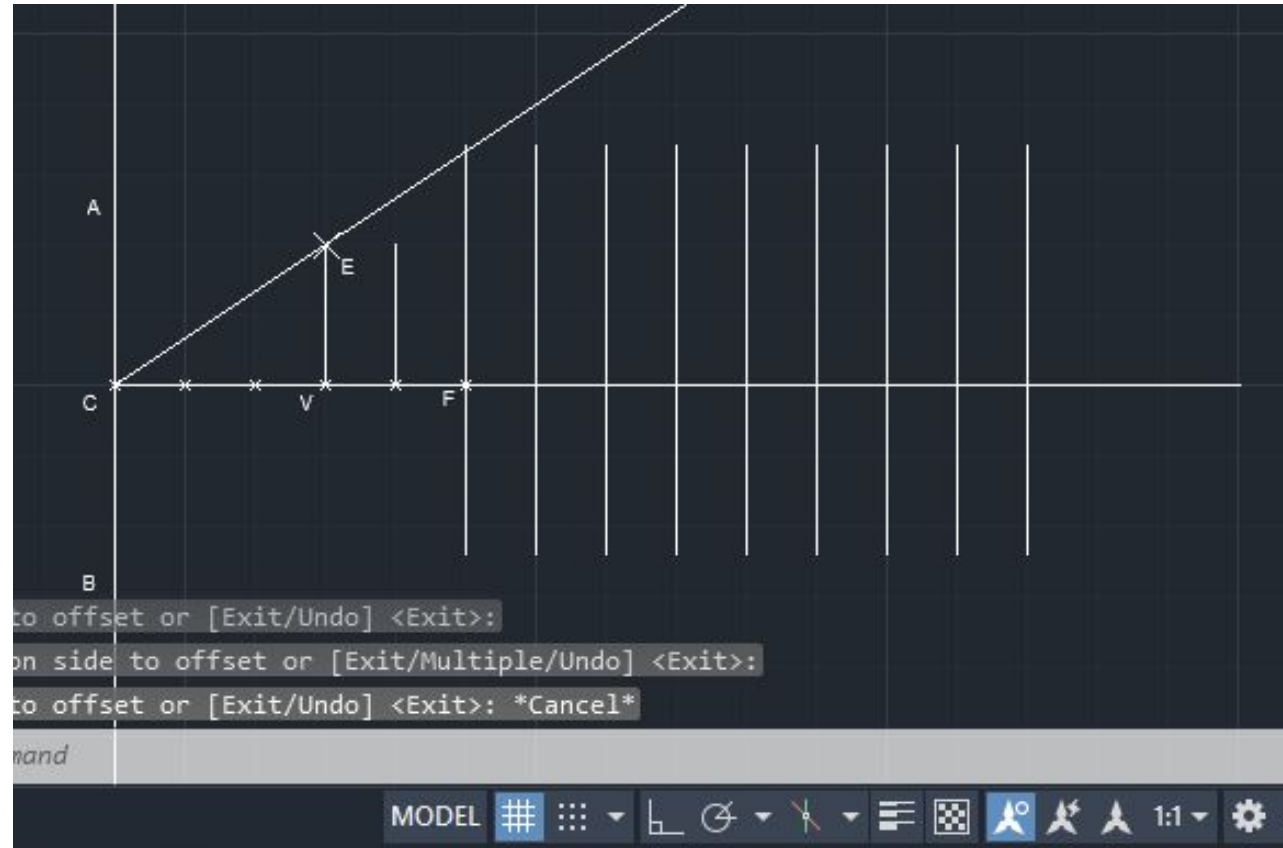
Note: Use the OFFSET command.

Keep selecting the line you want to create the OFFSET of and keep moving to the right.



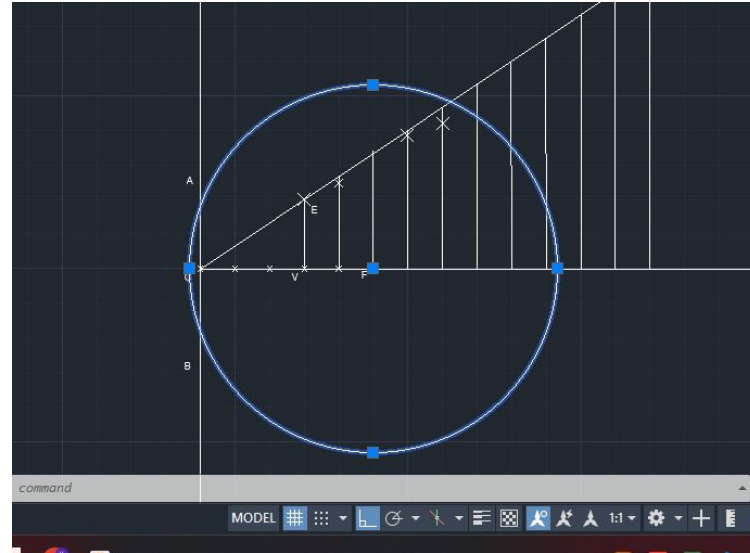
Ellipse using Focus Directrix Method:

Step 7: Then extend the line so that they intersect with CE.



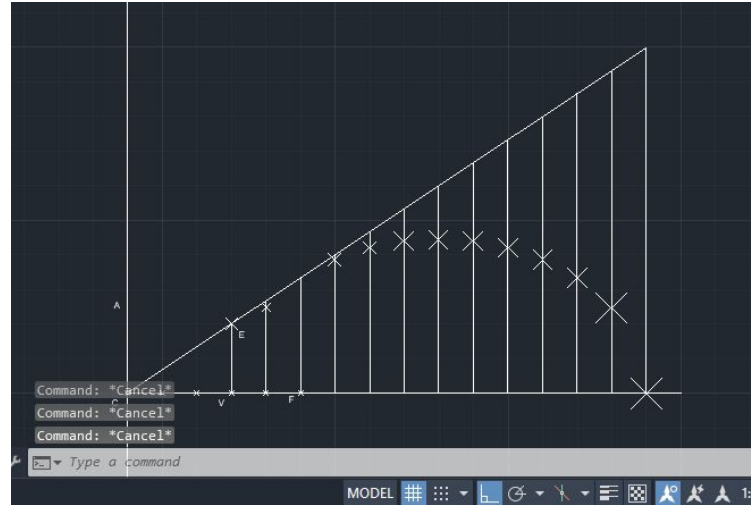
Ellipse using Focus Directrix Method:

Step 8: Draw a Circle keeping the radius equal to the distance from the horizontal line to the intersection of line CE. Now move the circle towards the focus keeping the centre of the circle at the focus. Then draw a point at the intersection of the vertical line with the circle.



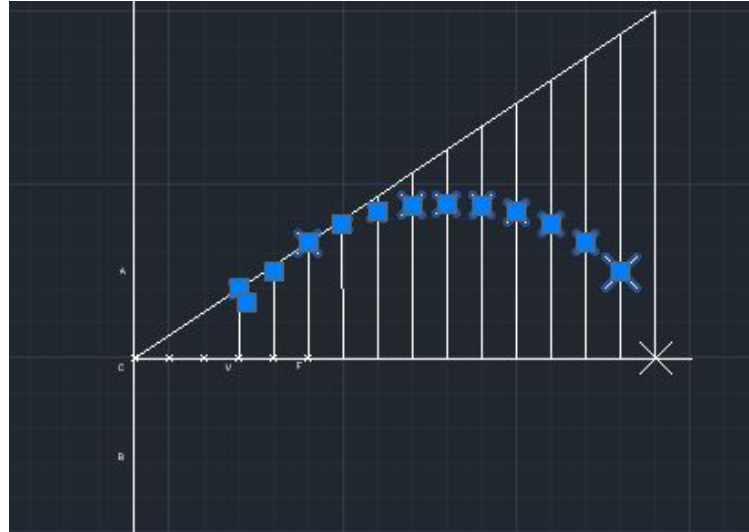
Ellipse using Focus Directrix Method:

Step 9: Repeat step 8 until the intersection point meets the horizontal line .



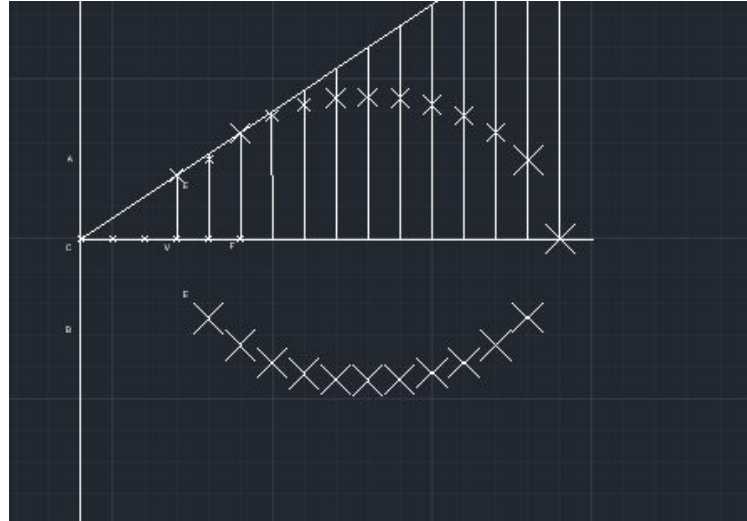
Ellipse using Focus Directrix Method:

Step 10: Select all the points.



Ellipse using Focus Directrix Method:

Step 11: Now mirror the above points with respect to the horizontal line.



Ellipse using Focus Directrix Method:

Step 12: SPLINE FIT line to connect all the points. Select CLOSE to close the line. This is the required ellipse.

